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# Analytical model of fully grouted bolts in pull-out tests and in situ rock masses

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ABSTRACT

The dynamic evolution characteristic of the bond strength at the interface of a bolt and a rock mass under an axial tensile load and the mechanical behavior of fully grouted bolts in situ were investigated considering the non-uniform stress of the surrounding rock along the bolts. A new dynamic bond-slip model was first proposed to describe the dynamic evolution characteristic of the bond strength at bolt-rock interfaces. Based on the proposed dynamic bond-slip model, analytical solutions of the shear stress distribution along the bolts, load-displacement relationship, and relative displacement considering the slip of the free end were developed. Then, incorporating the non-uniform normal stress of the surrounding rock along the bolts, the shear stress distribution and load-displacement behavior of the fully grouted bolts were presented. Moreover, analytical approaches were also proposed for determining the stress distribution along the bolt and rock deformation in cases of in situ pre-stressed and non-prestressed bolts. The proposed model and analytical solution were validated by published results from laboratory model tests and in situ tests, respectively. The new analytical solution completes the theoretical framework for addressing the fundamental problem of fully grouted bolts in pull-out tests and in situ rock masses and provides a useful theoretical tool that can potentially be applied to bolt design and laboratory and in situ testing.

## 1. Introduction

In recent years, rock bolts have been extensively used as an effective method in the reinforcement of tunnels and mining engineering projects. In these excavation projects, when a rock bolt is installed in a borehole drilled in a rock mass, the bolt-rock interaction at the interface tends to restrain the deformation and looseness of the strata surrounding the excavation, making the rock bolt a crucial component in tunnel reinforcement systems. However, the bond characteristics of the rock-bolt interface are still not fully formulated. To optimize the reinforcement effect of the bolt, it is essential to carry out an investigation of the mechanical behavior of the bolt.

Among the various types of rock bolts, fully grouted bolts are the most common in practical applications owing to not only its simple installation and outstanding reinforcement performance but also its low cost. A fully grouted bolt refers to a bolt that is inserted and grouted in a borehole along the entire length. Windsor[1] defined that an integral reinforcement system consists of four fundamental elements: the rock, the reinforcing element, the external fixture and the internal fixture. For a fully grouted bolt, the reinforcing element and the external fixture represent the steel bar and the faceplate and nut, respectively. The bonding effect between the bolt and rock mainly depends on the internal fixture, which refers to the grout material, for instance, the resin or cement mortar for grouted bars. In accordance with different grouting materials, fully grouted bolts can be divided into two major types: cement grouted bolts and resin grouted bolts. Since grouting technique and curing time of these two types of bolts are different, the formation mechanisms of grout strength of them are different, too. This research focuses on the cement grouted rock bolts.

A series of investigations were conducted to study the load transfer characteristic of the fully grouted bolts. In general, two mainstream methods are used to study the bond behavior of a bolt: (1) pull-out testing of the bolt attached to separated strain gauges and (2) field monitoring in the tunnel. Farmer[2] and Li and Stillborg[3] conducted pioneering works in pull-out testing and proposed corresponding mathematical models that predict an exponential decay of the shear stress to the free end. However, these early models have limitations, especially when the applied load is relatively high.

The difficulty of establishing a precise model is due to the complicated components of the interfacial bond strength. The generation of bond stress is probably not tied to a single factor but is a consequence of the combined action of the adhesion, mechanical interlock and axial

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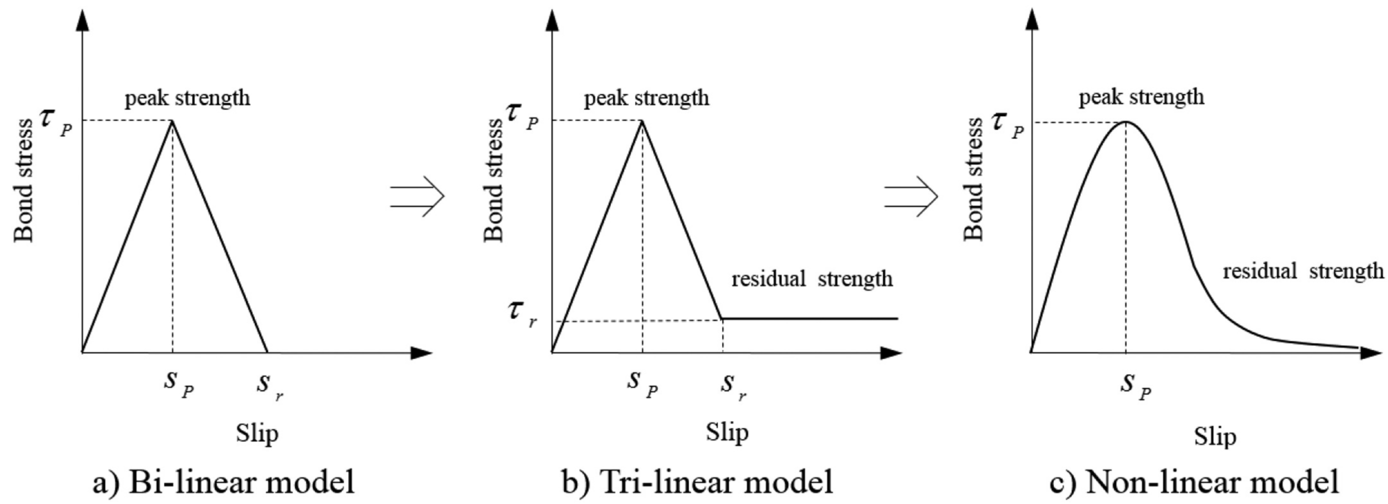


**Fig. 1.** Evolution of the bond-slip model.

friction. The problem can be dramatically simplified if the bond stress is controlled by a single parameter. A bi-linear or tri-linear bond-slip constitutive model was introduced (Fig. 1), from which a series of closed-form solutions for fully grouted bolts were derived.[4,5] Nevertheless, to some extent, the linear bond-slip model is unrealistic because experimentation shows that the bond stress and the relative slip are non-linearly related. The non-linear bond-slip model was put forward by Ma et al.[6] as depicted in Fig. 1. However, the aforementioned models are idealizations; Liu et al.[7] found that the bond-slip relationship varies along the bolt at different depths, indicating that the bond characteristic for fully grouted bolts in pull-out tests needs to be further studied.

Rock bolts in situ are widely used to reinforce the tunnel wall and unstable tunnel face[8] during the excavation process. In situ cases are distinct from pull-out tests. Freeman[9] carried out early field monitoring of bolts in the Kielder experimental tunnel and proposed the neutral point theory, which suggests that two shear stresses in opposite directions are generated at both sides of the neutral point. One segment of the bolt with a load dragging the bolt towards the excavation boundary refers to the pick-up length, while the other segment with a load pushing the bolt deep into the rock mass refers to the anchor length. A site monitoring study was conducted by Björnfot and Stephansson[10] (Fig. 2), validating Freeman's[9] theory. Cai et al.[11–13] presented analytical models for rock bolts in situ based on the convergence-confinement method by taking into account the complicated rock-bolt interactions.

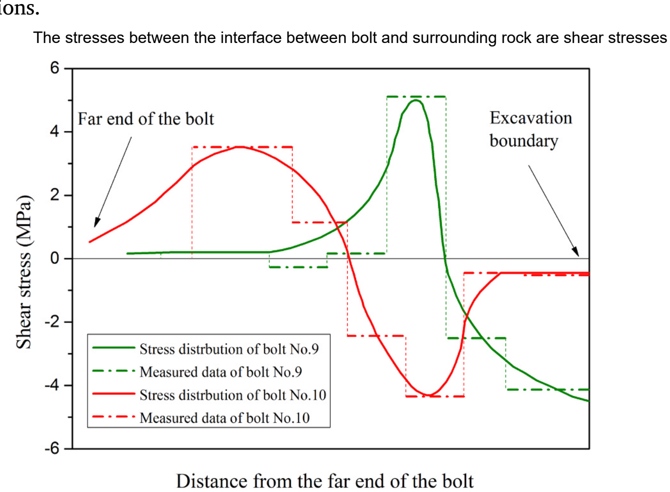


**Fig. 2.** Measured data (horizontal lines) and sketches of shear stress distributions along bolts No. 9 and No. 10, measured by Björnfot and Stephansson[10] in the field.

In addition to the analytical method, a series of numerical methods were also proposed for the in situ bolts to avoid the challenge of directly solving the non-linear differential equations, and some of these methods were implemented into discontinuous deformation analysis (DDA) algorithms for application to more complex scenarios.[14–17]

In conclusion, several obstacles exist in the analysis of fully grouted bolts. First, the practical bond-slip characteristic is hard to capture, and the previous models are idealizations. Second, the complicated interactions among the rock, grout, and bolt often require models with numerous parameters and formulas, which is inconvenient in application. Finally, the non-uniform deformation of the in situ rock is still not explicitly understood. This study aims to address the abovementioned issues and proposes simple, reliable, and accurate analytical models for fully grouted bolts in pull-out tests and in situ conditions.

## 2. Analytical model for bolts in pull-out tests

Pull-out testing is a conventional method to investigate the load transfer mechanism of fully grouted bolts. From these laboratory tests, researchers study the effect of the frictional resistance of bolts, improving the design and construction of in situ bolts. Generally, laboratory tests are preferred due to their lower cost and improved control for parametric studies, though such tests can be conducted either in situ or in a laboratory. The basic procedure is as follows: First, a bolt of thread steel (or smooth steel) is inserted into a rock sample (or a concrete block) by drilling a borehole, and then the bolt is encapsulated with grouting material (such as resin or cement). Once the installation is finished, a tensile force is applied at the end of the free section of the bolt. In this case, each bolt is instrumented with several axially placed electrical resistance strain gauges on the interface of the steel bar. During the loading process, the steel bar is loaded at a constant rate, and the variation in strain is monitored at each level.

### 2.1. Potential failure pattern of bolts

The load capacity of a rock bolt depends mainly on the failure patterns of the reinforcement system. It is essential to understand the potential patterns before analyzing the mechanical behavior of the bolt. For fully grouted rock bolts, there are several common failure modes: (a) If the strength of the bolt body is low, the bolt body may yield first; (b) If the strength of the rock mass is relatively high, or the bonding effect between the bolt and the mortar is weak, the decoupling may occur at the bolt-grout interface; (c) If the rock mass is relatively weak, the whole anchorage body (bolt and grout) may be pulled out together, which implies decoupling take place at the bolt-grout interface; (d) If

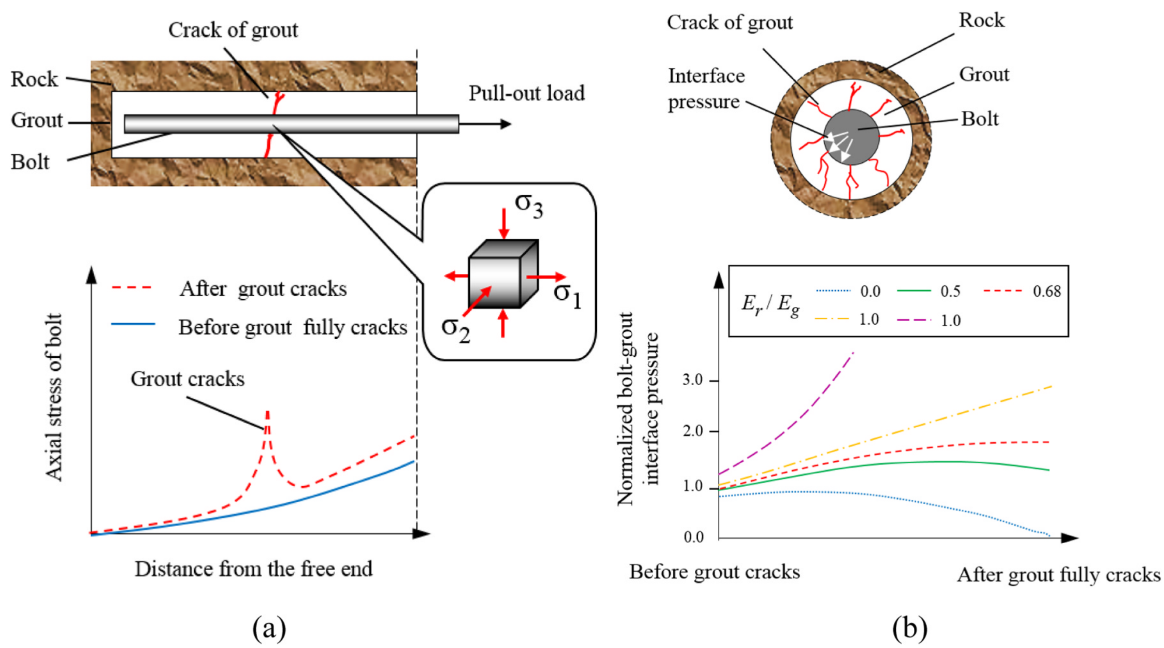


**Fig. 3.** Schematic diagram of stress concentration on the bolt when the grout cracks. (a) Concentration of axial stress at the crack of the grout; (b) The evolution of bolt-grout interface pressure at the crack of the grout.[19]

the rock is much weaker than the mortar and bolt, the unravelling of rock may happen.

These conventional failure modes are based on the condition that grout works effectively between bolt and the rock mass. If the grout is softer than required, it will easily fail and will not recover, so the anchoring effect of the anchor cannot be effectively utilized. On the other hand, if the grout is too hard, it will crack and the bolt may fail improperly due to the stress concentration. Li[18] pointed out that the premature bolt failure due to stress concentrations is often caused by the fracture or the joint opening. During the loading process of a grouted bolt, the crack may initiate at the bolt-grout interface and propagate radially towards the grout-rock interface.[19] Taking a crack plane as an example, a schematic diagram of the axial stress concentration is shown in Fig. 3(a). Before the grout cracks, the axial load declines exponentially towards the free end (The detailed analytical solutions will be introduced in following sections). In this situation, the bolt and grout bear the applied load together and the axial stress at a specific position can be expressed as:

$$\sigma_a = \frac{P_x}{A} \tag{1}$$

where $P_x$ and $A$ represents the load and effective section of the anchorage at a specific position. After the grout cracks, the grout at the crack plane will lose the ability to transmit load. Thus, the effective section will decrease significantly and the axial stress will concentrate on the bolt body, which leads to a peak value in Fig. 3(a). Yazici and Kaiser[19] studied the bolt-grout interface pressure during the crack propagation. In Yazici and Kaiser's[19] investigation, the radial pressure build-up at the bolt-grout interface was considered to be induced by the bolt dilatational movement against the grout or rock. Yazici and Kaiser[19] studied the evolution of bolt-grout interface pressure during the cracking process and the results is shown in Fig. 3(b). In Fig. 3(b), the horizontal axis of the coordinate represents the process of crack propagation, and the vertical axis represents the ratio of the bolt-grout interface stress to the grout shear strength. For the grout that is not too strong ($E_r/E_g > 0.68$), the cracks should grow in a stable manner[19] and the interface pressure will continue to increase. However, if the grout is very hard ($E_r/E_g < 0.68$), it should suddenly be propagating in an unstable manner[19], and the interface pressure may increase first and then decrease. $E_r$ and $E_g$ represents the Young's modulus of the rock and the grout. In Fig. 3(a), the element in crack plane is taken, we have:

$$\begin{cases} \sigma_1 = \sigma_a \\ \sigma_2 = \sigma_3 = \sigma_r \end{cases} \tag{2}$$

where $\sigma_a$ and $\sigma_r$ represents the axial stress and bolt-grout radial stress at the crack plane. According to Tresca yield criterion, maximum shear stress in the material can be calculated as:

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} \tag{3}$$

In 19, the value of the bolt-grout interface pressure is found to be relatively low (0–20 MPa). In the case of low applied load, the axial stress of the bolt at the crack is also relatively low (say less than 50 MPa), and the maximum shear stress may not increase. But when the applied load is relatively high, the maximum shear stress will increase significantly and the premature failure may take place at the crack plane.

On the basis of laboratory and field investigations, Kilic et al.[20] suggested the slippage at the bolt-grout interface as the most dominant failure pattern. However, rock bolts are mainly preferred at weak rock mass conditions. Therefore, together with grout-rock bolt interface, rock-grout interface may be crucial as well. Out of the failure modes previously specified, our discussion will concentrate on the yielding of the bolt, decoupling at the bolt–grout or grout–rock interface.

### 2.2. Dynamic non-linear bond-slip model

For fully grouted rock bolts installed in a rock mass, once subjected to an axial tensile load, shear stress at the bolt-grout interface (or grout-rock interface) will be generated to resist the external load, as shown in Fig. 4. Since the 1970s, numerous researchers have studied the generation and distribution of shear stress along rock bolts and proposed a range of models to describe the mechanical behavior of the rock bolt under tension. For example, Li and Stillborg[3] proposed a model consisting of a piecewise function to predict this distribution along the bolt, as shown in Fig. 4. The bond stress distribution under different levels of applied load was calculated by Li and Stillborg[3]; in this work, the bolt was considered to be divided into several sections under different

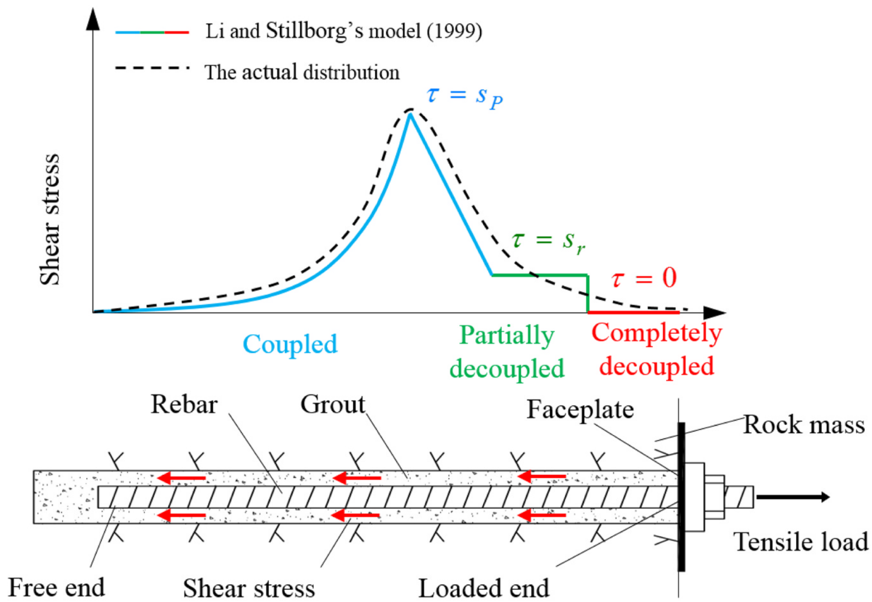


**Fig. 4.** Schematic diagrams of a fully grouted rock and the distribution of shear stress along the bolt-grout interface under an axial tensile load.

incompatible deformation: the grout and the bar deformed differently and slip occurs as they are decoupled

loading stages. These sections can be defined as completely decoupled, partially decoupled, or coupled. As Li and Stillborg[3] postulated, in the section of the bolt that undergoes incompatible deformation, the bolt-grout interface (or grout-rock interface) is completely or partially decoupled with zero shear stress and a constant residual shear stress, while in the section that undergoes compatible deformation, the bond stress linearly increases to the peak strength and then exponentially attenuates to the free end of the steel bar. (The end under tension is defined as “loaded end” and the other end as “free end” as shown in Fig. 4) However, this assumption is limited in practical applications; for example, the length of the linearly ascending segment must be assumed in advance to obtain the shear stress expression, which may cause problems during calculation. In addition, partial and completely decoupled parts are not constant in the actual situation, as the load on the bolt shank increases, the coupling mechanism must be changing. Benmokrane et al.[21] proposed a classical tri-linear bond-slip model for establishing the interfacial constitutive relationship. As depicted in Fig. 1, the piecewise function $\tau(S)$ comprises a linearly increasing segment to the point representing the peak shear strength $\tau_p$, followed by a decreasing section representing the softening region, and a horizontal line segment characterizing the residual strength $\tau_r$. This type of model was later commonly adopted by other researchers. Ren et al.[4] used the abovementioned conventional bond-slip model to derive a closed-form solution for the full-range behavior of a rock bolt by taking into account five loading stages. In the model of Ren et al.[4], the loading stage of the bolt must be known before the parameters can be substituted into the corresponding equation to calculate the shear stress distribution; this methodology may be inconvenient. In 2011, the model of Ren et al.[4] was modified by Martin et al.[5] using a refined tri-linear bond-slip model, which focuses on the behavior of the free extremity of the bolt and allows the use of a unique and crucial parameter to characterize the interfacial behavior. In this conventional tri-linear model, the relationship between the bond stress $\tau$ and relative displacement $\Delta s$ is generally assumed to be $\tau = k\Delta s$, where $k$ is the interfacial tangential stiffness. However, the coefficient $k$ is difficult to determine because the shear stress distribution along the bolt is non-uniform and non-linear, according to the results of pull-out testing.

To obtain a better understanding of the interfacial mechanical behavior, it is essential to investigate the non-linear constitutive relations between the rock bolt and the grout medium. Dai et al.[22] and Zhou et al.[23] performed pioneering works on the bond-slip relationship for externally-bonded fiber reinforced polymer (EB-FRP) joints and derived analytical solutions for the stress distribution along the strip under an axial pull-out load. Ma et al.[6] proposed a non-linear bond-slip model for a fully grouted rock bolt based on the method used by Zhou et al.[23] and verified their model with two short encapsulation pull-out tests. In the Ma et al.'s[6] model, the $\tau - s$ constitutive relations were assumed to be invariable along the embedment length of the bolt; nonetheless, the discrepancy of the bond-slip property at different locations has been ignored. Liu et al.[24] verified this characteristic through laboratory experimentation and proposed a dynamic broken line model, which is a numerical calculation model, to compute the ultimate bearing capacity using an iteration method. Liu et al.[24] demonstrated that the value of the interfacial peak bond stress tends to increase as the location $x$ moves away from the loading point. In Liu et al.'s[24] model, the ultimate shear stress and the critical depth at which the peak bond stress begins to increase are not yet elucidated. Thus, to address the issue mentioned above and derive a more precise solution for the stress distribution along a bolt, it is essential to propose a new dynamic $\tau - s$ constitutive model.

### 2.2.1. *Basic assumptions*

According to the previous discussion, the $\tau - s$ constitutive model is used to describe the bonding mechanism at the bolt-grout (or grout-rock) interface. To avoid ambiguity, the following derivation is based on the assumption that the relatively slip occurs at the bolt-grout interface, and the case of grout-rock interface will be discussed later.

### 2.2.2. *Strain and slip distribution*

Fig. 5 shows the calculation model of rock (or grout)-bolt relative slip. According to 22, if Young's modulus of the bolt $E_b$ and the radius of the bolt $r_b$ are known, the relationship between the strain of the reinforcing element and the relative slip of the bolt-grout interface can be calculated as follows.

$$\varepsilon(x) = f[S(x)] \tag{4}$$

where $\varepsilon(x)$ and $S(x)$ are the strain and slip at any location $x$ and $x$ is the distance from the free end of the bolt.

Based on various laboratory experiments in which different conditions are adopted, a unique form of expression of $f(S)$ can be obtained

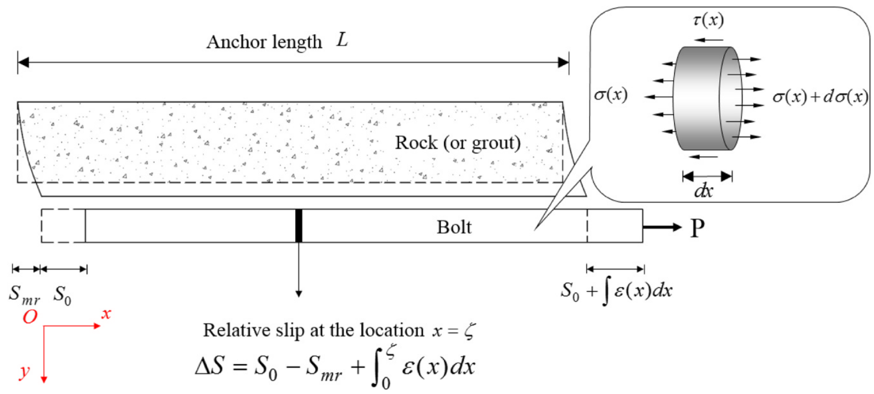


**Fig. 5.** Calculation model of fully grouted bolts subject to a tensile load.

as follows.

$$f(S) = A(1 - e^{-BS}) \tag{5}$$

where $A$ and $B$ are the coefficients related to the pull-out test.

For the conditions that this study has assumed, the relative slip $S(x)$ can be calculated as the sum of the displacement at the free end of the bolt $S_0$, the micro-deformation of the rock mass $S_{mr}$, and the integration of the strains from the free end to the considered point $x$, as shown in Fig. 5. However, since the micro-deformation of the rock sample around the bolt is hard to be observed. For simplification of analysis, the micro-deformation $S_{mr}$ in the pull-out test is assumed to be zero, so the expression can be written as

$$S(x) = S_0 + \int_0^x \varepsilon(x)dx \tag{6}$$

Rewriting Eq. (6), we have

$$\varepsilon(x) = \frac{dS(x)}{dx} \tag{7}$$

Substituting Eqs. (4) and (5) into Eq. (7), the differential equation can be expressed as

$$\frac{dS(x)}{dx} = A(1 - e^{-BS(x)}) \tag{8}$$

Considering the following boundary conditions,

$$\varepsilon(L) = \frac{\sigma(L)}{E_b} = \frac{P}{E_b \pi r_b^2} \tag{9}$$

where $L$ represents the anchor length of the bolt.

The *relative displacement* $S(x)$ *between the bolt and the rock can be obtained by solving Eq.* (8):

$$S(x) = \frac{1}{B}\ln(1 + e^{AB(x-\mu)}) \tag{10}$$

where $\mu$ is a parameter related to the applied load $P$, and $\mu$ can be expressed by

$$\mu = L + \frac{1}{AB}\ln\left(\frac{AE_b\pi r_b^2}{P} - 1\right) \tag{11}$$

It should be noted that the model of slip distribution proposed in this study is initially identical to that developed in 22 and 23, although the deducing process is different. In Dai et al.'s[22] investigation, the $\varepsilon$~$s$ relationship was measured at the loading end of the bolt under the boundary conditions of zero strain and zero displacement at the free end ($\varepsilon(0) = 0$, $S(0) = 0$); however, in a practical situation, the free end slip $S_0$ is a parameter related to the pull-out load ($\varepsilon(0) = 0$, $S(0) \neq 0$). Additionally, the influence of embedment length $L$ in the traditional $\varepsilon$~$s$ model was ignored, precluding the expression of the variation in the $\tau$~$s$ relationship. With the aim of meeting the boundary condition and taking into account the effect of $L$, the expressions for the strain and slip distribution should be modified.

Substituting Eq. (10) into Eq. (7) gives

$$\varepsilon(x) = A\left(\frac{1}{1 + e^{-AB(x-\mu)}}\right) \tag{12}$$

In Eq. (12), the strain asymptotically approaches zero as the location approaches $x = 0$. At the free end, the Eq. (12) gives $\varepsilon(0) = A\left(\frac{1}{1+e^{AB\mu}}\right) \neq 0$, which does not agree with the condition mentioned above; therefore, Eq. (12) can be refined as follows.

$$\varepsilon(x) = A\left(\frac{x/L}{x/L + e^{-AB(x-\mu)}}\right) \tag{13}$$

which agrees well with the condition $\varepsilon(0) = 0$ and takes into account the effect of $L$. Adding Eq. (13) into Eq. (6) gives

$$S(x) = S_0(\mu) + \int_0^x A\left(\frac{x/L}{x/L + e^{-AB(x-\mu)}}\right)dx \tag{14}$$

in which the free end slip $S_0(\mu)$ is subject to the applied load $P$ and is hard to be measured. Therefore, the free end slip $S_0(\mu)$ can be approximated by substituting $x = 0$ into Eq. (10) as

$$S_0(\mu) = \frac{1}{B}\ln(1 + e^{-AB\mu}) \tag{15}$$

The value of Eq. (15) varies with the pull-out load $P$. Therefore, Eq. (10) is modified to the approximate equation for $S(x)$:

$$S(x) = \frac{1}{B}\ln(1 + e^{-AB\mu}) + \int_0^x A\left(\frac{x/L}{x/L + e^{-AB(x-\mu)}}\right)dx \tag{16}$$

### 2.2.3. $\tau$~$S$~$x$ relationship

Section 2.2.2 indicates that once the parameters $A$, $B$ and $\mu$ are obtained, $S(x)$ and $\varepsilon(x)$ can also be determined. The next step is to investigate the relationship between the shear stress $\tau$ and strain $\varepsilon$ or slip $S$. As is depicted in Fig. 5, on the basis of the balance of an infinitesimal element of the bolting system, the equilibrium equation is expressed as

$$(\sigma(x) + d\sigma(x) - \sigma(x))\cdot\pi\cdot r_b^2 = \tau(x)\cdot 2\cdot\pi\cdot r_b\cdot dx \quad \text{i.e. ,} \frac{d\sigma(x)}{dx} = \frac{2\tau(x)}{r_b} \tag{17}$$

According to Hooke's law, Eq. (17) can be rewritten as

$$E\frac{d\varepsilon(x)}{dx} = \frac{2\tau(x)}{r_b} \tag{18}$$

Combining Eqs. (13), (16) and (18) gives the $\tau$~$S$~$x$ bond-slip relationship:

$$\begin{cases} \tau = \frac{Er_b}{2} \cdot A^2 B \cdot \frac{x + 1/AB}{L} \cdot \frac{e^{-AB(x-\mu)}}{[x/L + e^{-AB(x-\mu)}]^2} \\ S = \frac{1}{B}\ln(1 + e^{-AB\mu}) + \int_0^x A\left(\frac{t/L}{t/L + e^{-AB(t-\mu)}}\right)dt \end{cases} \tag{19}$$

It should be noted that the parameter $\mu$ can be eliminated in Eq. (19) mathematically. Thus, the dynamic non-linear bond-slip model is a function independent of $P$, but about $S$ and $x$.

At weak rock mass conditions, the failure may mainly occur at the grout-rock interface. In such case, the proposed approach is also valid, just by using an equivalent Young's modulus $E_e$ of the anchorage body and a borehole radius$r_e$, which can be expressed as:

$$\begin{cases} r_e = r_g \\ E_e = \frac{E_b r_b{}^2 + E_g(r_g{}^2 - r_b{}^2)}{r_g^2} \end{cases} \tag{20}$$

where $r_e$ and $E_e$ represent the equivalent radius and material Young's modulus; $E_b$ and $E_g$ represent the Young's modulus of bolt and grout; $r_b$ and $r_g$ represent the radius of bolt and borehole.

Compared with the previous approach, this model describes the bond-slip relationship with a binary function $\tau(S, x)$, which is independent of the load level $P$ and bond length $L$, namely, the $\tau$~$S$ relationship varies with location. This phenomenon is distinct from those in traditional models, in which the shear-slip response curve is constant along the bond length of the rock bolt.

In Eq. (19), the parameters $A$ and $B$ correlate with the property of the grout-bolt interface; these two parameters can be uniquely fixed with a regression analysis method. This method can be implemented by using MATLAB's curve fitting toolbox to fit the strain distribution data, which can be measured by strain gauges bonded along the bolt. The goodness of fitting is represented by the coefficient "R-square", which can be calculated as follows :

$$R - square = 1 - \frac{\sum_{i=1}^{n}(y_i - y_0)^2}{\sum_{i=1}^{n}(y_i - \bar{y}_0)^2} \tag{21}$$

where $y_i$ is the fitting data, $y_0$ and $\bar{y}_0$ are actual value and mean value of the measured data. Thus, the closer "R-square" comes to 1, the better the curve coincides with the measured data. The parameters can be obtained with a higher accuracy when a greater amount of strain data under various loading levels are used.

Liu et al.[24] conducted an anti-pullout anchor experiment of fully grouted bolts encapsulated in a mixture to investigate the load transfer properties of anchorage body at different depths. The sketch of experimental set up and the essential parameters are shown in Fig. 6(a). The axial strain of the anchor under different loads is measured by the strain gauges, and the shear stress at different depths can be calculated as

$$\tau_i = \frac{Er_b(\varepsilon_{i+1} - \varepsilon_i)}{2l} \tag{22}$$

where $l$ is the gauge separation, $r_b$ is the radius of the bolt, and $\varepsilon_i$ is the $i$th strain data.

Moreover, since only the slip of the loaded end can be directly measured, the bolt slip at different depths can be obtained by subtracting the deformation of the bolt from the measured slip according to Fig. 5:

$$S_j = S_L - \sum_{i=j+1}^{4}\frac{(\varepsilon_{i+1} + \varepsilon_i)}{2}l_i - \frac{(\varepsilon_j + \varepsilon_{j+1})}{4}l_j \tag{23}$$

where $S_L$ is the loaded end slip, $S_j$ is the $j$th slip, $l_j$ is the $j$th gauge separation, and $\varepsilon_i$ is the $i$th strain data. In the test, the strength of the mixture is low so the decoupling takes place first at the grout-mixture interface. In this situation, the equivalent Young's modulus of the anchorage body calculated as 33.5 GPa. Using Eq. (13) to fit the measured strain data, parameters $A$ and $B$ for the dynamic non-linear model are obtained as 0.00021 and 19, respectively. As a comparison, the non-linear $\tau$~$S$ model proposed by Ma et al.[6] is also presented in this section. The parameters needed in Ma et al.'s model[6] are adopted as: ultimate load $P_{max}$ = 2.23kN, peak elastic slip $S_e$ = 0.058mm, initial tangential stiffness $K_0$ = 38.42kN/mm. The comparison results are shown in Fig. 6(b) and the experimental verification is also shown in Fig. 6(b). As depicted in Fig. 6(b), the work of Liu et al.[24] shows that the evolution of shear stress with displacement is different at different depths; meanwhile, the dynamic model proposed in this work agrees well with the laboratory data. For comparison, a deviation exists between the measured value and the curve calculated by the model of 6 One probable reason causing this deviation might be that the bond length $L$ is quite short, and the traditional bond-slip model is applicable to bolts only with relatively long embedded lengths. Hence, the new function $\tau(S, x)$ might be more reliable for modeling the $\tau$~$S$ response at any location along a bolt with any possible bond length. In Fig. 6(c), the three curves represent the bond-slip relationships at different locations, and the loading stages can be divided into four general stages. Take the $\tau$~$S$ model in Fig. 6(c) as an example, in the elastic stage, the critical displacement is $S_S$ and the ultimate bond stress is $\tau_S$. When the slip is $S < S_S$, similar linear relationships are found between the shear stress and interfacial displacement at different depths. Then, the $\tau$~$S$ relations tend to separate, and the bond stress continues to increase to the peak stress $\tau_{pk}$ during the plastic-hardening stage. Additionally, $\tau_{pk}$ increases with depth within the rock mass. While $S > S_{pk}$, the shear stress approximately decreases with the exponent level, revealing that the interface is under a plastic-softening state and that decoupling is considered to occur when $\tau$ approaches zero.

According to the phenomenon described above, one physical interpretation of the dynamic bond-slip relationship could be the change in boundary condition. The surface deformation of the rock near the free end might be restrained by the intact rock block behind it, unlike the deformation in areas away from the deep section of borehole. In other words, the shear stress required for the unit deformation near the free end is more than that adjacent to the loaded end.

### 2.3. Bolt behavior in pull-out tests

In Section 2, the dynamic bond-slip model was established; if the relative displacement is obtained, the stress distribution can be calculated directly. For pull-out tests, substituting Eq. (13) into Eq. (18) yields

$$\tau(x) = \frac{Er_b}{2} \cdot A^2 B \cdot \frac{x + 1/AB}{L} \cdot \frac{e^{-AB(x-\mu)}}{[x/L + e^{-AB(x-\mu)}]^2} \tag{24}$$

It should be observed that the distribution of shear stress is affected by the bond length $L$. When $L \to \infty$, the shear stress and axial strain are considered to be generated near the loading point, and the free end slip $S_0$ can be assumed to be negligible, mathematically expressed as follows: $x \to \infty$, $x/L \to 1$, $\tau(x) \to \frac{Er_b}{2} \cdot A^2 B \cdot \frac{e^{-AB(x-\mu)}}{[1 + e^{-AB(x-\mu)}]^2}$. The result closely resembles the solution of Ma et al.[6], which is applied to bolts with long bond lengths and zero slip at their free ends. Combining Eq. (16) with Eq. (13) $\mu$ gives

$$\begin{cases} \varepsilon = A\left(\frac{x/L}{x/L + e^{-AB(x-\mu)}}\right) \\ S = \frac{1}{B}\ln(1 + e^{-AB\mu}) + \int_0^x A\left(\frac{t/L}{t/L + e^{-AB(t-\mu)}}\right)dt \end{cases} \tag{25}$$

Considering $P(S) = A \cdot \sigma(S) = AE \cdot \varepsilon(S) = E\pi r_b^2 \cdot \varepsilon(S)$, the load-displacement expression can be rewritten as

$$\begin{cases} P = EA\pi r_b^2\left(\frac{x/L}{x/L + e^{-AB(x-\mu)}}\right) \\ S = \frac{1}{B}\ln(1 + e^{-AB\mu}) + \int_0^x A\left(\frac{t/L}{t/L + e^{-AB(t-\mu)}}\right)dt \end{cases} \tag{26}$$

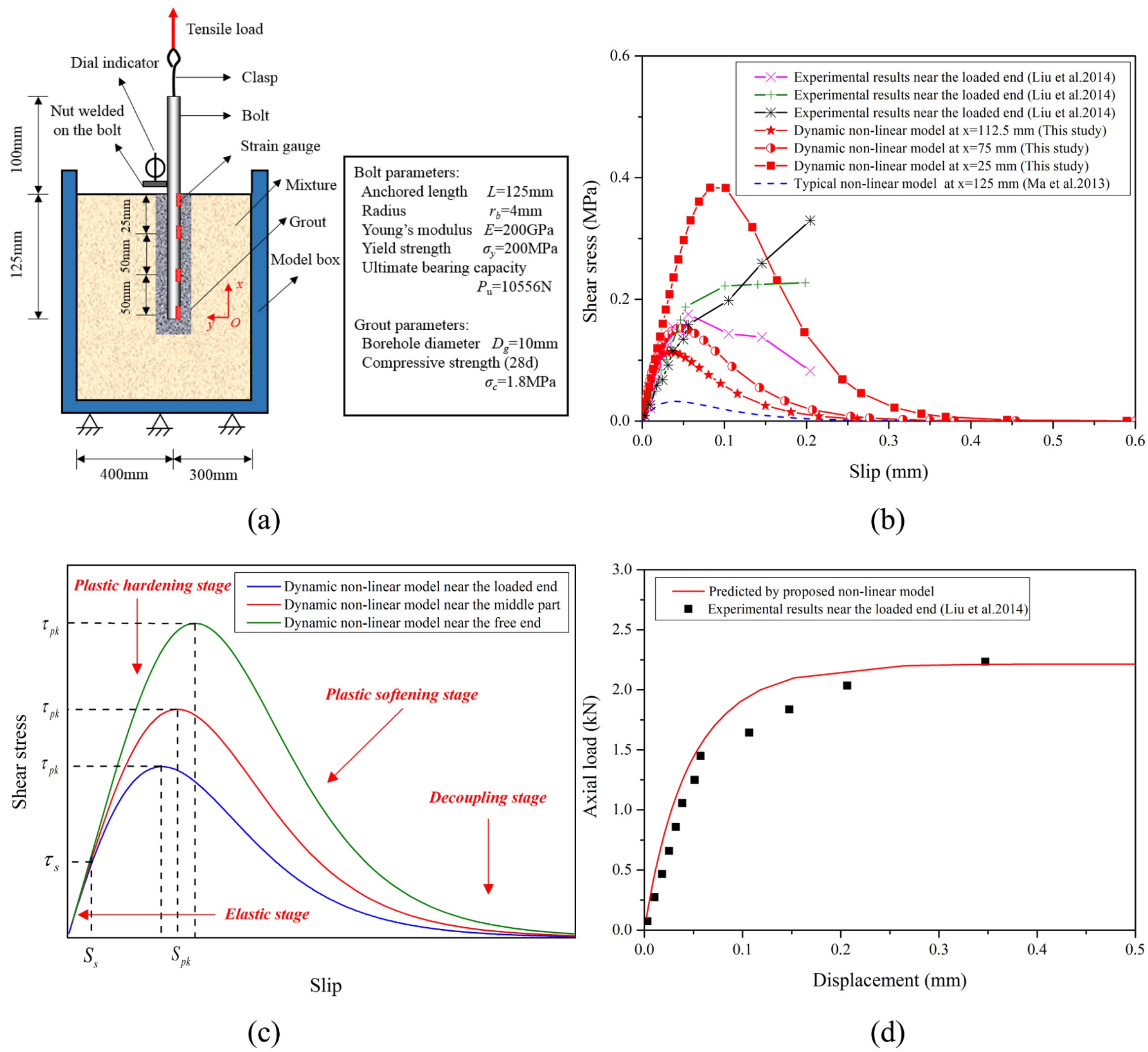


**Fig. 6.** Analytical solutions for specimen M02# in the test of Liu et al.[24] (a) A sketch of experimental set up in the test of 24 (b) Comparison of bond-slip models for specimen M02# in the test of 24 (c) A schematic of loading stage in the dynamic bond-slip model. (d) Comparison of the load-displacement curve for specimen M02# at x = $L$ in the test of 24.

Unlike the model proposed by Ma et al.[6], which first needs to be calibrated by the measured $P-S$ curve, Eq. (26) provides a direct approach to predict the load-displacement relationship, which requires several sets of strain gauge readings. In the example analyzed above, Liu et al.[24] recorded the data of the $P-S$ relationship, and these data are plotted in Fig. 6(d). The results predicted by Eq. (26) are also shown as a solid line in Fig. 6(d), which clearly shows that the model calculated by Eq. (26) accurately reproduces the $P-S$ curve during the loading process. The predicted ultimate load is 2.21 kN and the measured value is 2.22 kN, and the relative error is 0.5%.

### 2.3.1. Experimental verification

Rong et al.[25] carried out pull-out tests of thread steel inserted in concrete to simulate and study the mechanical behavior of bolt-rock interface as shown in Fig. 7(a). Strain gauges were installed along the axial direction of the bolt at 50–100 mm intervals to monitor the axial strain. The parameters for the pull-out test as also listed in Fig. 7(a). The coefficients $A$ and $B$ needed for the proposed model are determined as 0.002 and 2.8 by using Eq. (13) to fit the strain gauge readings in Fig. 7(b), and the correlation coefficient is 0.99. The analytical predictions of the strain distribution, shear stress distribution, load-displacement relation and dynamic bond-slip model are calculated with Eq. (13), Eq. (24), Eq. (26) and Eq. (19). In addition, these predictions are plotted in Fig. 7(b)-(d), in which the test results are also included for comparison.

As shown in Fig. 7(b), when the steel bar was subjected to different levels of applied loads, the presented model agrees well with the measured data. The coefficients $A$ and $B$ in Fig. 7(b) are determined by several groups of strain data from approximately the elastic stage (the definition of each stage is ambiguous in non-linear models), where stable strain gauge readings can be obtained easily. Then, $A$ and $B$ are used to predict the full-range behavior of the bolt.

Fig. 7(c) illustrates the contrast between test results (discrete points) and the analytical prediction (lines). Fig. 7(c) shows a close match between the measured data and the analytical model. Additionally, the bond stress increases towards the free end of the bolt.

In Fig. 7(d), the load-displacement curve predicted by the proposed model is indicated by a solid line, while the load-displacement data measured in the laboratory test are plotted as discrete points. The solid line shows an accurate prediction of the full-range behavior of the bolt, which means that the prediction is dependable. However, the ultimate load of the prediction is about 337 kN, while the bolt in the test was measured to yield at around 300 kN. The cause of this error is the yield strength of the thread steel is 390 MPa, that is to say, the theoretical yield load is $P_y = \pi r_b^2 \sigma_b = 314$ kN. This means failure has been occurred in the bolt body while the bolt-concrete interface is still well coupled,

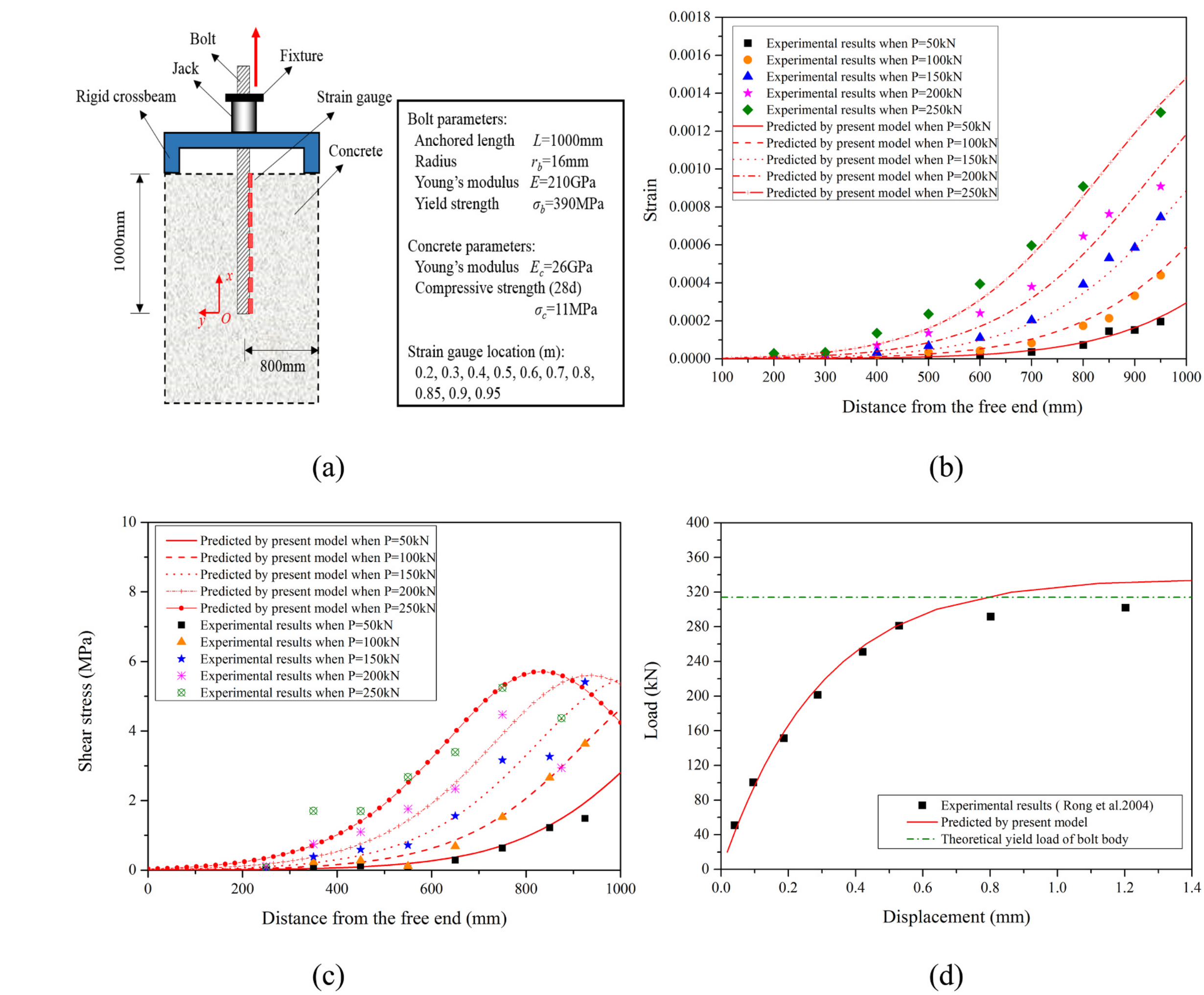


**Fig. 7.** Analytical solutions for the Rong et al.[25] test. (a) A sketch of experimental set up in the test of 25. (b) Comparison of test results and model predictions of the strain distribution for the[25] test. (c) Comparison of test results and model predictions of the shear stress distribution for the[25] test. (d) Comparison of test results and model predictions of the load vs. slip at the loaded end for the[25] test.

which is consistent with the experimental observation.

In addition to the non-linear model of Martin et al.[5], the tri-linear bond-slip models proposed by Ren et al.[4] and Martin et al.[5] are also added to the comparison in this section. Fig. 8 clearly shows that the Martin et al.[5] method and Ma et al.[6] non-linear model produce similar results, which closely agree with the dynamic non-linear model at the

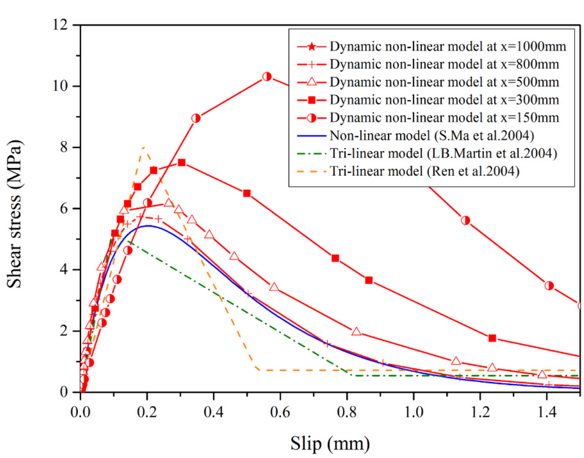


**Fig. 8.** Comparison between several typical bond-slip models and the proposed approach.

loaded end in this study. Moreover, the model proposed in this study reflects that the $\tau - S$ relationship varies with the change in location along the bolt, which is consistent with the discussion in Section 2.

### 2.4. Full-range analysis of the interfacial behavior

In Section 2.3, the proposed model in this study has been verified by practical engineering applications; however, the issue still remains that the readings of the measuring instruments are often unstable when decoupling occurs, impeding the analysis of the full-range behavior of the bolt. Using the analytical model developed above, the mechanical behavior along the steel bar during the whole loading process can be obtained as long as the parameters $A$ and $B$ are calibrated with strain data in the stability region (in the elastic stage or softening stage). Fig. 9(a) and (b) show the evolution of the axial load and shear stress along the bolt from the beginning of the test to complete failure.

As illustrated in Fig. 9(b), the loading process is divided into three stages, which are plotted as lines of different color. When the pullout force is relatively small, the whole bolt is in the elastic state, and no decoupling is considered to exist at the interface. The bond stress at $x = L$ increases constantly until it reaches the peak stress ($\tau(S, L)$ in Eq. (19)). In addition, the shear stress at other positions exponentially decrease to zero, similar to the assumptions used in previous studies.

As the applied load continues to increase, softening occurs at $x = L$, and the bond stress begins to decrease; in contrast, the bond stress in

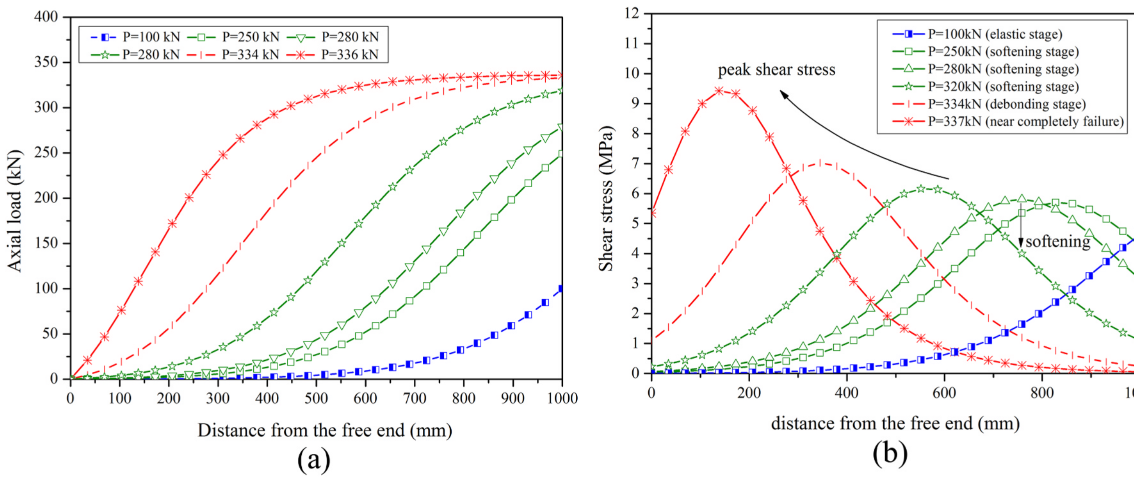


**Fig. 9.** Full-range behavior of the fully-grouted bolt under tensile load. (a) Full-range behavior of the axial load along the bolt. (b) Full-range behavior of the shear stress along the bolt.

the regions not far behind the loaded end increases owing to plastic hardening at the interface, causing a continuously increasing peak stress towards the embedded end. That is, the bond stress at each specific position first increases and then decreases during the loading process. This behavior implies that the bar undergoes a softening stage.

When the pull-out force approaches 335 kN, the bond stress at $x = L$ approximately approaches zero, as depicted in Fig. 9(b), which marks the onset of the debonding stage. In this stage, a complete decoupling occurs at the loaded point and propagates towards the free end. When the pull force increases slightly, the decoupling front rapidly moves to the free end. The proposed model indicates that the segments near the free end have a higher peak stress from this stage, which can be explained by theoretical explanation that the grout-bolt interface near the extremity needs to mobilize more bond stress to prevent the steel bar from being pulled out under a growing load. This phenomenon is consistent with the Liu et al.'s[24] test results. When the load $P$ exceeds 336 kN, the interfacial shear stress reaches the ultimate value and can no longer balance the external load, leading to a complete decoupling between the concrete and the reinforcing element.

A mathematical approach can also be introduced to explain the variation characteristic of the peak stress. For a specific location $x = x^*$ on the bolt, taking the derivative of Eq. (24) with respect to $x^*$, and letting it be zero gives

$$\mu = x^* + \frac{1}{AB}\ln\left(\frac{2 + 2ABx^* + A^2B^2(x^*)^2}{ABL(2 + ABx^*)}\right) \tag{27}$$

According to Eq. (11), $\mu$ is a parameter related to the applied load $P$. Eq. (27) describes the relationship between the position where the shear stress peak appears and the external load $P$. In Eq. (11), when $P$ approaches the ultimate load, $\left(\frac{AE_b\pi r_b^2}{P} - 1\right) \to 0$. On the basis of the property of a natural logarithm, a drastic change occurs in the value of $\mu$. Thus, the stability coefficient can be defined as follows.

$$\phi = \left(\frac{AE_b\pi r_b^2}{P} - 1\right) \tag{28}$$

It is assumed that the value of the peak stress is stable when $|\phi| \geq 0.01$. Substituting the values of $A$, $E$ and $r_b$ into Eq. (28) gives $P \leq 334$ kN. This result implies that the value of the peak stress is stable (approximately invariable) until the load exceeds 334 kN, which agrees with the earlier discussion. Then, substituting $P = 334$ kN into Eq. (11) gives $\mu = 200$, then solving Eq. (27), we have $x^*=347$ mm. The peak shear stress is nearly constant when $347 \leq x \leq 1000$ mm (stable segment); however, the peak value changes greatly when $0 \leq x \leq 347$ mm (unstable segment). According to this characteristic, the supporting performance of the fully grouted bolt might be improved by enhancing the bond strength of the unstable segment in the design of the bolt.

## 3. Analytical model for rock bolts in situ

In tunnel construction, rock bolts are often used to restrain the deformation of the rock mass. After tunnel excavation, deformation of the rock mass gradually increases with time; therefore, the rock bolt is subjected to a non-uniform load along the bond length due to interfacial shearing. For a circular tunnel, unreinforced rock mass tends to move towards the excavation boundary, and the displacement of the exposed rock is greater than that of the interior rock. The rock bolt deforms with the rock mass after it is installed in the borehole. Unlike in the pull-out test, in situ, the displacement of the rock towards the tunnel wall exceeds that of the bolt in the zone near the opening; however, the situation is reversed in the deeper region of the surrounding rock. This phenomenon results in shear stress acting in two directions, as depicted in Fig. 10. Since 1978, several field investigations have been conducted to study the stress distribution of in situ rock bolts. Freeman[9] carried out field monitoring of the fully grouted bolts in the Kielder experimental tunnel and summarized the characteristics of the interfacial behavior. In that study, the concepts of the neutral point, pick-up length and anchor length were first proposed. At the neutral point, the bolt deforms consistently with the rock mass, causing the shear stress to be zero and the axial load to reach a peak value at this point. Moreover, along the pick-up length, the shear stress is oriented towards the opening, while the direction of the shear stress reverses towards the free end of the bolt, as shown in Fig. 10. This result implies that the difference between the bolts in the pull-out test and in the field is that the in situ bolt has a pick-up length and anchor length, but the experimental bolt has only an anchor length.

### 3.1. Analytical solution for bolts in situ

Based on the theory of the bond-slip relationship previously discussed, the bond stress at the bolt surface is determined by the relative displacement at the bolt-grout interface. For simplicity, the rock mass and the grout are assumed to be whole; therefore, the relative displacement in the following derivation is considered to occur at the bolt-grout interface only. The final deformation of the rock and bolt can be interpreted to occur in two steps, as shown in Fig. 11 (a) and (b). For the first step, the deformation is considered to be consistent across the rock-bolt interface, i.e., the reinforcement of the bolt is ignored in this step. $U_{r0}$ represents the free displacement of the rock mass without

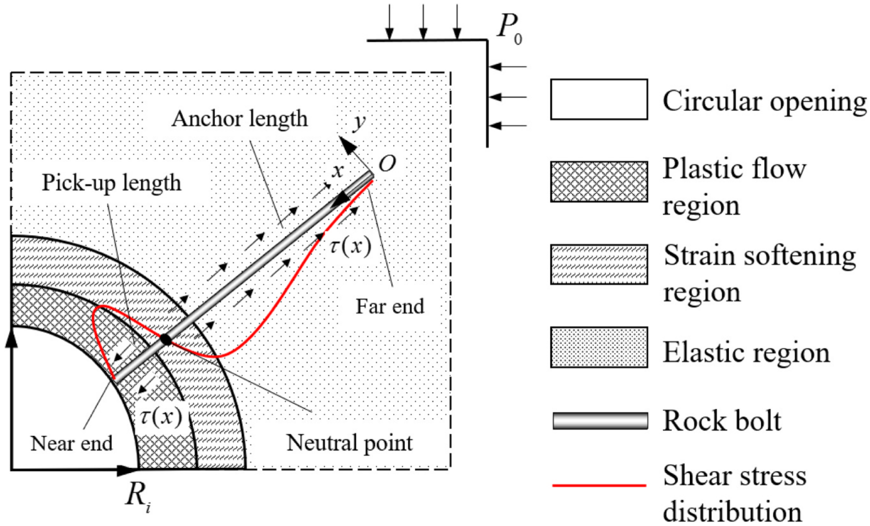


**Fig. 10.** Shear stress distribution for rock bolts in situ.

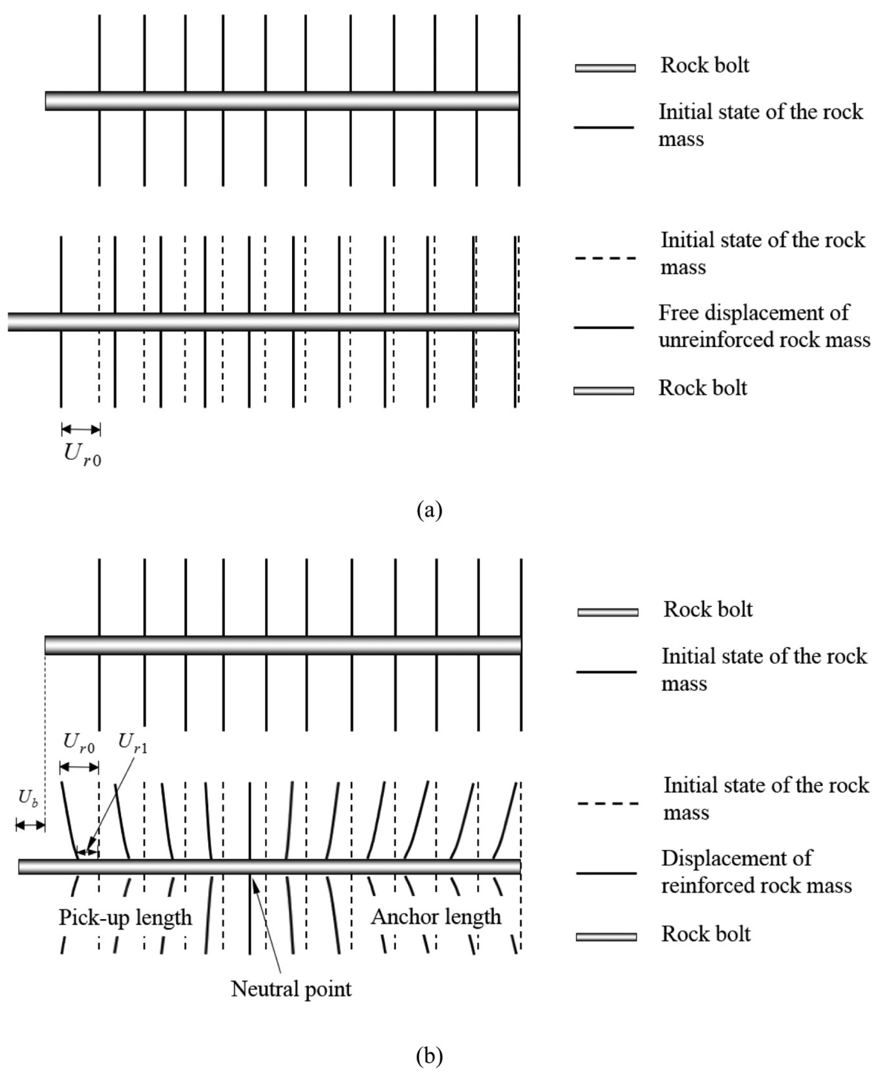


**Fig. 11.** Displacement of a bolt and its surrounding rock mass in the two steps. (a) Displacement of the rock mass, ignoring reinforcement. (b) Displacement of the rock mass, considering reinforcement.

bolting (Fig. 11 (a)). In the second step, the bolt reinforcement is into the rock mass. For a single rock bolt, the axial displacement of the rock mass near the bolt is restrained after it is bolted. The rock displacement at the bolt surface is $U_{r1}$, which will be discussed next.

According the bond-slip relationship mentioned above, the shear stress at the interface is determined by the relative displacement, which can be expressed as

$$\Delta U = U_{r1} - U_b \tag{29}$$

where $U_{r1}$ is the displacement of the rock at the bolt surface with bolting and $U_b$ is the displacement of the bolt.

Determination of displacements $U_{r1}$ and $U_b$ is important for the analysis of the behavior of the bolting system. Steel bars either with or without a face plate are two common types of rock bolts. For the bolt with a face plate, a portion of the force generated at the bolt-grout interface is transferred to the face plate, resulting in a non-zero axial stress at the end near the tunnel wall. However, free displacement of the rock mass occurs in the case without a face plate. In other words, the corresponding boundary conditions for these two situations are slightly different. In this section, two closed-form solutions with different boundary conditions are derived for fully grouted bolts in the field.

### 3.1.1. Bolts without a faceplate

For an in situ bolt without a faceplate, the two ends of the bolt are free; therefore, the axial strain at these two positions are zero, which can be written as

$$\begin{cases} \varepsilon(0) = 0 \\ \varepsilon(L) = 0 \end{cases} \tag{30}$$

in the coordinate system shown in Fig. 10.

Numerous field monitoring results show that the axial load reaches the peak value at the neutral point and exponentially attenuates to the far end of the bolt, away from the tunnel wall. (The end installed at the excavation boundary is defined as "near end" and the other end is defined as "far end", as shown in Fig. 10) Thus, the axial strain of the bolt can be assumed to be

$$\varepsilon(x) = \frac{\alpha x(L - x)}{e^{L+\beta x}} \tag{31}$$

where $\alpha$ and $\beta$ are coefficients to be determined and $L$ is the bond length of the bolt.

Considering that the slip at the far end of the bolt still remains unclear and that the absolute value of this slip is relatively small in this case, the total displacement of the bolt can be calculated as follows.

$$S(x) = \int_0^x \varepsilon(x) = \alpha\frac{e^{-L-\beta x}[\beta^2 x(x - L) - \beta(L - 2x) + 2]}{\beta^3} + \theta \tag{32}$$

where $\alpha$, $\beta$, and $\theta$ are coefficients to be determined.

To determine the values of $\alpha$, $\beta$, and $\theta$, more boundary conditions should be taken into account. According to Eq. (18), the shear stress at the neutral point ($x = \xi$) can be written as

$$\tau(\xi) = \frac{Er_b}{2}\varepsilon'(\xi) \tag{33}$$

where $x = \xi$ denotes the position of the neutral point.

From the neutral point theory, we have

$$\tau(\xi) = 0 \tag{34}$$

By adding Eq. (34) into Eq. (33), we get

$$\beta = \frac{2\xi - L}{\xi^2 - L\xi} \tag{35}$$

Moreover, the coefficients $\alpha$ and $\theta$ should satisfy the following conditions

$$\begin{cases} \alpha \cdot S(L) + \theta = U_{b0}(L) \\ \alpha \cdot S(\xi) + \theta = U_{r0}(\xi) \end{cases} \tag{36}$$

where $U_{r0}(x)$ and $U_{b0}(x)$ denote the free displacement of the rock mass and the final displacement of the bolt, respectively. $U_{r0}(x)$ can be predicted mathematically, and $U_{b0}(L)$ can be directly measured from the tunnel wall.

Accordingly, the axial load distribution along the bolt can be calculated directly as follows.

$$\sigma(x) = \pi r_b^2\frac{\alpha x(L - x)}{e^{L+\beta x}} \tag{37}$$

where $\alpha$ and $\beta$ are determined from Eq. (35) and Eq. (36).

### 3.1.2. Bolts with a faceplate

Compared to the case mentioned above, the rock bolt with a face plate has distinct boundary conditions; thus, the stress partly transfers to the face plate. In particular, when decoupling occurs, the released rock deformation will influence the plate, eventually leading to a non-zero value of the axial load at the excavation boundary. To fit the characteristics of the bolt in this case, the axial strain along the bond length can be obtained by rewriting Eq. (31) as

$$\varepsilon(x) = \frac{\alpha x(L + \omega - x)}{e^{L+\beta x}} \tag{38}$$

where $\alpha$, $\beta$, and $\omega$ are coefficients to be determined and $L$ is the bond length of the bolt. $\omega$ represents the offset of the neutral point and can be expressed as

$$\omega = \xi - \xi_0 \tag{39}$$

where $\xi$ and $\xi_0$ denote the positions of the neutral point with and without a face plate, respectively.

Similarly, combining Eq. (33), Eq. (34) and Eq. (38) gives

$$\beta = \frac{2\xi - (L + \omega)}{\xi^2 - (L + \omega)\xi} \tag{40}$$

Thus, the axial load distribution in the case of a bolt with a face plate can be expressed as

$$\sigma(x) = \pi r_b^2\frac{\alpha x(L + \omega - x)}{e^{L+\beta x}} \tag{41}$$

where $\alpha$, $\beta$, and $\omega$ are determined by Eq. (38), Eq. (39) and Eq. (40).

When the distribution of the axial load is obtained, the resultant shear stress at the bolt surface can be directly calculated using the differential relations provided in Eq. (18).

## 3.2. Application example for a rock bolt in situ

In this section, an example from the Holland Zaka Tunnel in Nagasaki is introduced to show the application of the proposed model for in situ bolts. In 2004, Cai et al.[13] used an analytical method based on an improved shear-lag model to predict the stress distribution in the case of the Holland Zaka Tunnel. This underground tunnel is located in volcanic deposits. According to 26 and 13, the ground pressure is relevant to released tunnel deformation which is influenced by bolt installation time and Geological conditions of the tunnel area. In this case, the released displacement is estimated as 40% of the total displacement and the other required parameters are shown as follows. Radius of the tunnel is $r_i$ = 4.75 m; Initial in-situ stress in the rock is $P_0$ = 1.0 MPa; Young's modulus of the rock mass is $E_m$ = 0.5 GPa; Poisson's ratio of the rock mass is $\nu$ = 0.35; Length of the rock bolt is $L$ = 4 MPa; Diameter of the bolt is $d_b$ = 25.4 mm; Young's modulus of the bolt is $E_b$ = 210 GPa; Young's modulus of the bolt (*assumed by author) is $E_g$ = 30 GPa; and Peak bond stress is $S_p$ = 1.2 MPa.

For a rock bolt without a face plate, the first step is to determine the free displacement of the rock mass $U_{r0}$. It is assumed that the tunnel is in

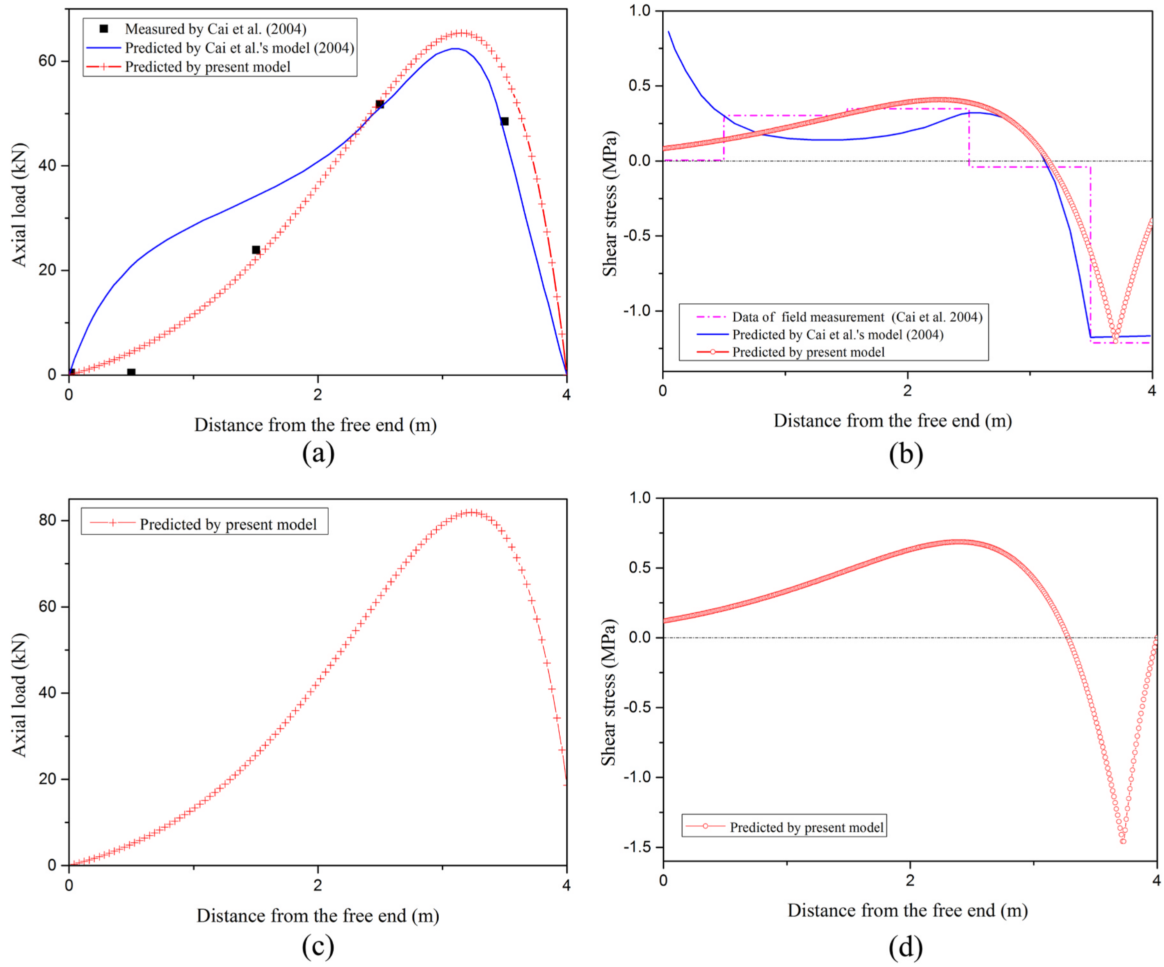


**Fig. 12.** Calculation results of in situ rock bolts (measured by 13). (a) Distribution of the axial load for a bolt without a face plate. (b) Distribution of the shear stress for a bolt without a face plate (The measured data is represented by the horizontal line) (c) Distribution of the axial load for a bolt with a face plate. (d) Distribution of the shear stress for a bolt with a face plate.

an elastic state which was subjected to a in-situ stress field $P_0$; thus, the distribution of the radial displacement in the rock mass can be expressed as

$$U_{r0}(x) = \frac{P_0(1+\nu)r_i^2}{[E_m(r_i+x)]} \tag{42}$$

This equation was proposed in 12. Moreover, if the rock bolt was considered to be installed across the plastic-flow region, plastic softening region and elastic region, the numerical method is preferred to calculate the displacement of the rock mass.[27,28] The next step is to obtain the bolt displacement at the tunnel wall; this value can be directly measured. In this case, the displacement of the rock bolt $U_{b0}$ is estimated to be 11.25 mm. The position of the neutral point is the last parameter needed to be determined. Neutral point theory has been discussed by many researchers. Freeman[9] first proposed the concept based on the result of field monitoring in the Kielder experimental tunnel. Then, Tao and Chen[29] derived an expression to calculate the position of the neutral point by assuming a linear relationship between the shear stress and relative displacement at the bolt-rock interface. However, in their solution, the position is related to only the radius of the tunnel $r_i$ and the length of the bolt $L$. Li and Stillborg[3] obtained an analytical solution for the interfacial behavior and suggested that the position of the neutral point depends on many factors. In brief, the neutral point can be determined with an empirical formula or analytical solutions. In this section, the Li and Stillborg[3] solution is used to calculate the location of the neutral point $\xi$; The result is $\xi = 3.15$, and the detailed expressions are shown as follows:

$$\xi = L + r_i - \lambda \tag{43}$$

where $r_i$ is the radius of the circular opening, $L$ is the embedment length of the rock bolt, and $\lambda$ is the distance from the center of the tunnel to the neutral point. $\lambda$ can be calculated via solving Eq. (44) (for bolts with a face plate) and Eq. (45) (for bolts without a face plate):

$$\gamma G_r\left(\frac{r_b}{2}\cdot\frac{d^2u}{d\lambda^2} - \frac{\psi}{2}\int_{r_i}^{\lambda}\frac{d^2u}{dt^2}e^{-\psi\frac{\lambda-t}{r_b}}dt\right) - \frac{\psi}{2}\gamma G_r\left(\int_{r_i}^{r_p}\frac{d^2u}{dt^2}dt\right)e^{-\psi\frac{\lambda-r_p}{r_b}} = 0 \tag{44}$$

$$\gamma G_r\left(\frac{r_b}{2}\cdot\frac{d^2u}{d\,\lambda^2} - \frac{\psi}{2}\int_{r_p}^{\lambda}\frac{d^2u}{dt^2}e^{-\psi\frac{\lambda-t}{r_b}}dt\right) + \frac{\psi}{r_b}s_r(r_p - r_i)e^{-\psi\frac{\lambda-r_p}{r_b}} = 0 \tag{45}$$

where $r_p$ can be obtained by solving Eq. (46) (for bolts with a face plate) and Eq. (47) (for bolts without a face plate):

$$\frac{s_p}{\gamma G_r} + \frac{r_b}{2}\cdot\frac{d^2u}{dr_p^2} - \frac{\psi}{2}\int_{r_i}^{r_p}\frac{d^2u}{dt^2}dt = 0 \tag{46}$$

$$s_p + \gamma G_r\frac{r_b}{2}\cdot\frac{d^2u}{dr_p^2} + \theta s_p\cdot\frac{\psi}{r_b}(r_p - r_i) = 0 \tag{47}$$

$$G_r = \frac{E_r}{2(1+\nu)},\ G_g = \frac{E_g}{2(1+\nu)} \tag{48}$$

$$\psi = \frac{2G_r G_g}{E_b\left[G_r \ln\left(\frac{r_g}{r_b}\right) + G_r \ln\left(\frac{r_0}{r_g}\right)\right]} \tag{49}$$

$$\gamma = \frac{2(1+\nu)SE_b}{\pi r_b^2 E_b + SE_m} \tag{50}$$

where $G_r$, $G_g$ are shear modulus of rock and bolt; $r_g$ is the radius of the grout and $r_0$ is the influencing radius of a single rock bolt; $S$ is the influencing area of a single rock bolt; $\nu$is the Poisson's ratio of the rock mass; $\theta$ is a parameter related to the residual strength of the bolt-grout interface; $\psi$is a parameter related to the properties of the rock, bolt and grout; The second-order derivative of the displacement $u$ of the rock can be expressed as

$$\frac{d^2u}{dx^2} = -\frac{P_0}{G_r}\cdot\frac{r_i^2}{x^3} \tag{51}$$

where $P_0$ is the hydrostatic primary stress in the rock mass.

According to Section 3.1, the final parameters needed for the application of the new analytical solution for a bolt without a faceplate are calculated as follows: $U_{r0}(\xi) = 10.88\text{mm}$, $U_{b0}(L) = 11.25\text{mm}$, $\xi = 3.15\text{m}$, $\alpha = 8.35 \times 10^{-4}$, $\beta = -0.86$, $\theta = 1.02 \times 10^{-2}$.

The distribution of axial load and shear stress are shown in Fig. 12(a) and (b), and data from field monitoring and the model of 13 are also shown in these figures.

When the shear stress reaches the peak stress near the tunnel wall, decoupling may occur. The debonding of in situ bolts is more complex than that in pull-out tests. Thus, for simplicity, it is assumed that the shear stress distribution of the decoupled section is symmetrical with that in the section from the location of the neutral point to the peak stress. As is shown in Fig. 12(a) and (b), the measured data and the analytical solution are consistent, which indicates that the proposed approach is reasonable.

For a bolt with a face plate, the application of the model is similar to that in the approach mentioned above, and the required parameters are shown as follows: $U_{r0}(\xi) = 11.44\text{mm}$, $U_{b0}(L) = 11.75\text{mm}$, $\xi = 3.28\text{m}$, $\alpha = 1.18 \times 10^{-3}$, $\beta = -0.87$, $\theta = 0.99 \times 10^{-3}$, $\omega = 0.13$.

The distributions of the axial load and shear stress are shown in Fig. 12(c) and (d).

The shear stress at the face plate is considered to be zero; thus, the shear stress in the decoupled section is assumed to decay exponentially to zero. Fig. 12(c) and (d) show that the bolt with a face plate has a non-zero load at the tunnel face and that the position of the neutral point is closer to the opening.

Furthermore, the radial deformation of the rock mass in the influencing area of a single rock bolt is not uniform, and the reinforcement effect gradually decays with distance away from the bolt. In reality, the relative displacement at the bolt surface is difficult to measure. The deformation of the rock mass at the bolt surface can be calculated by combining the bond-slip model with the in situ model. Taking the case without a face plate as an example, the bond-slip relationship can be assumed, as shown in Fig. 13(a). It should be mentioned that this case is more complicated than pull-out tests, so an approximate bond-slip model is used here

$$\tau(S, x) \approx \tau(S, L) \tag{52}$$

Combining the assumed bond-slip relationship and the shear distribution obtained above (Fig. 12(b)), the relative displacement $\Delta U$ can be determined. Then, the rock displacement at the bolt surface can be calculated as

$$U_{r1}(x) = U_{b0}(x) - \Delta U = \alpha\frac{e^{-L-\beta x}[\beta^2 x(x-L) - \beta(L-2x) + 2]}{\beta^3} + \theta - \Delta U \tag{53}$$

The results of three typical displacement distributions for the case of a bolt without a faceplate are plotted in Fig. 13(b). The deformations of the rock and bolt are very similar from the far end of the bolt to the position near the peak stress, indicating that both the rock and the bolt couple well at the bolt surface. However, from the position of the peak stress to the tunnel wall, the deformations are not consistent, which implies that decoupling has occurred. With the proposed approach, once the bond-slip relationship is determined by the pull-out test, the interaction behavior at the rock-bolt interface can be explicitly modeled.

## 4. Analytical model for pretensioned rock bolts

Pretensioning of grouted rock bolts is a widely used method in the field. For cement-grouted bolts, the pretension should be applied after the grout becomes cured.[30] This means that the interaction between bolt and rock cannot be neglected in the process of applying prestress. Ranjbarnia et al.[31] proposed a simplified model to compute the stress distribution along the pretensioned bolts by assuming that the rock and the bolt are rigidly connected (no shearing displacements exist between the rock and the bolt). Furthermore, Ranjbarnia et al.[32] also indicated that the bolt head stiffness (deformation characteristics of nut, washer, and the plate) may have a dramatic impact on the performance of the pretensioned bolts. Hence, in an effort to analyze the load distribution of the grouted bolts at installation stage and working stage, a new

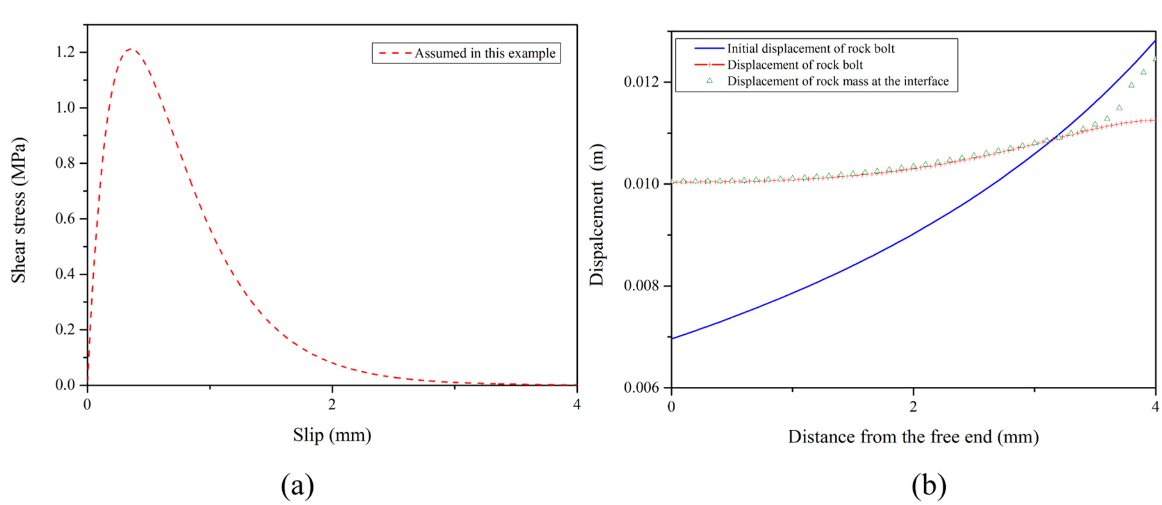


**Fig. 13.** Typical displacement distributions for the case of a bolt without a faceplate. (a) Assumed bond-slip relationship for the case without a face plate. (b) Typical displacement distribution for the case without a faceplate.

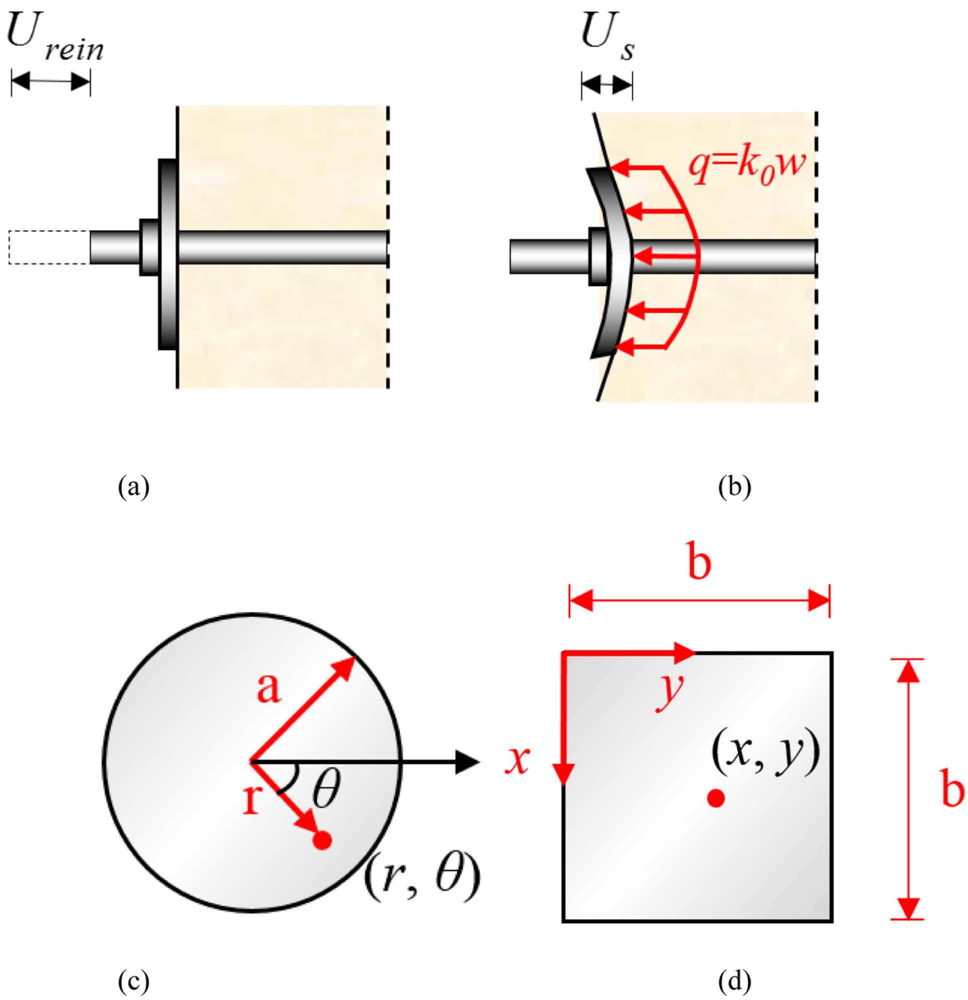


**Fig. 14.** Calculation model for pretensioned rock bolts. (a) A sketch of the deformation of the rock and the bolt (step one). (b) A sketch of the deformation of the rock and the bolt (step two). (c) A sketch of circular bolt plate. (d) A sketch of square bolt plate.

approach is presented in this section by considering the shearing displacement on the bolt interface and the influence of plate deformation.

In practical situation, the force actually applied to the tunnel surface $P_s$ is less than the initial pretensioning load $P_{ini}$. Ranjbarnia et al.[32] attributed this to the deformation of the bolt head and the compression of the rock mass. Thus, the actual force on the tunnel surface can be expressed as[32]:

$$P_s = \varphi P_{ini} \tag{54}$$

where $\varphi$ is a parameter which associates to the stiffness of the bolt head and reinforcing element. However, the method to determine specific value of $\varphi$ is not presented. In this section, the process of applying pretension can be divided into two steps (Fig. 14(a),(b)). For the first step, the initial pre-tensioned force causes the axial elongation of the bolt $S_{ini}$, which is analogous to the case of pull-out tests in Section 2. For the second step, a portion of the bolt elongation is reduced because the deflection of the face plate $w$ is induced by the applied force $P_{ini}$. Subsequently, the redistribution of the stress along the bolt occurs and the load on the bolt head drops to $P_s$. Eventually, the displacement on the bolt head can be expressed as:

$$S_s = S_{ini} - w \tag{55}$$

The shape of face plate is either flat or domed.[30] For a domed plate, the dome can accommodate a portion of deformation caused by partial external force. Thus, the deflection of the face plate can be calculated as:

$$w = \frac{P_{ini}}{K_d} \tag{56}$$

where $K_d$ is the equivalent stiffness of the domed plate. For a simply flat plate, it can be seen as a thin plate with free edges on elastic foundations. The reaction force at any point of the plate is assumed to be proportional to the deflection $w$, which can be expressed as:

$$q = K_0 w \tag{57}$$

where $K_0$ is the coefficient of rock reaction. Since the exact solution of the plate under such an assumption is complex, approximate approaches are introduced to calculate the value of $w$. For a circular plate (Fig. 14(c)), when a pretensioning load $P_{ini}$ acts at the center of the plate, the total potential energy of the system can be obtained as[33,34]:

$$U = \int_0^{2\pi}\int_0^{r_a}\left[\frac{D}{2}\left\{\left(\frac{\partial^2 w}{\partial r^2} + \frac{1}{r}\frac{\partial w}{\partial r} + \frac{1}{r^2}\frac{\partial^2 w}{\partial \theta^2}\right)^2 - 2(1-\nu)\left[\left(\frac{\partial^2 w}{\partial r^2}\right)\left(\frac{1}{r}\frac{\partial w}{\partial r} + \frac{1}{r^2}\frac{\partial^2 w}{\partial \theta^2}\right) - \left(\frac{1}{r}\frac{\partial^2 w}{\partial r\partial\theta} - \frac{1}{r^2}\frac{\partial w}{\partial \theta}\right)^2\right]\right\} - q(r,\theta)w + K_0\frac{w^2}{2}\right]rdrd\theta \tag{58}$$

where $r_a$ is the radius of the plate, $(r, \theta)$ is the coordinates of a particular point in Fig. 14(c), $D$ is the bending stiffness of the plate, which can be expressed as:

$$D = \frac{Eh^2}{12(1-v^2)} \tag{59}$$

where $E$ and $v$ represent the Young's modulus and Poisson's ratio of the plate, respectively; $h$ is the thickness of the plate. Based on the general solution of circular plate, an approximate function of deflection is selected[33,34]:

$$w = A + B\left(\frac{r}{r_a}\right)^2 \tag{60}$$

where $A$ and $B$ are parameter to be determined. Substituting Eq. (60) into Eq. (58), gives:

$$U = \frac{4\pi D}{{r_a}^2}(1+v)B^2 + \pi K_0\left(\frac{1}{2}A^2{r_a}^2 + \frac{1}{2}AB{r_a}^2 + \frac{1}{6}B^2{r_a}^2\right) - P_{ini}A \tag{61}$$

If the system is in a position of stable equilibrium, the total energy is a minimum. Minimizing with respect to $A$ and $B$, we have:

$$\begin{cases} A + B\left(\frac{2}{3} + \frac{16D(1+v)}{K_0{r_a}^4}\right) = 0 \\ A + \frac{B}{2} = \frac{P_{ini}}{\pi K_0{r_a}^2} \end{cases} \tag{62}$$

Solving Eq. (62) gives the function of deflection:

$$w = \frac{P_{ini}}{\pi[K_0{r_a}^4 + 96D(1+v)]}\left(\frac{4K_0{r_a}^4 + 96D(1+v)}{K_0{r_a}^2} - 6r^2\right) \tag{63}$$

When $r = 0$, the above function represents the deflection at the center of the plate. For a square plate (Fig. 14(d)), Tan[35] chose another function of deflection to obtain the approximate solution of deflection at the center point:

$$w_O = M\left(\frac{2+\pi^2+3v}{v}\right) + N \tag{64}$$

$$M = \frac{60P_{ini}{l_b}^2v(6+\pi^2+9v)}{1440D\pi^4(6+5v+4v^2) + K_0{l_b}^4[8\pi^4 + 45(20+24v+5v^2)]} \tag{65}$$

$$N = -\frac{P_{ini}\{-1440D\pi^4(6+5v+4v^2) + K_0{l_b}^4[32\pi^4 + 120\pi^2(2+3v) - 45(20+24v+5v^2)]\}}{K_0{l_b}^2\{1440D\pi^4(6+5v+4v^2) + K_0{l_b}^4[8\pi^4 + 45(20+24v+5v^2)]\}} \tag{66}$$

where $l_b$ is the side length of the square.

In order to simplify the subsequent derivation, it is assumed that the bolt-grout interface is in an elastic state when the bolt is subjected to initial prestension. According to Section 2, the shear stress increases approximately linearly with the relative displacement. So the shear stress can be expressed as:

$$\tau = K_{ini}\Delta S \tag{67}$$

where $K_{ini}$ is the initial shear stiffness between the bolt and grout.[32] Substituting Eq. (4) and Eq. (67) into Eq. (7) gives:

$$\frac{d^2S}{dS^2} = \delta^2 S \tag{68}$$

where

$$\delta = \sqrt{\frac{2K_{ini}}{E_b r_b}} \tag{69}$$

As mentioned above, the initial prestress firstly leads to the elongation of the bolt, the bolt displacement at the tunnel surface can be obtained by substituting the following boundary conditions into Eq. (68):

$$\begin{cases} S'(L) = \varepsilon(L) = \frac{P_{ini}}{E_b\pi r_b^2} \\ S'(0) = \varepsilon(0) = 0 \end{cases} \tag{70}$$

Thus, the solution of bolt displacement at the first step is obtained as follows:

$$S_{ini}(x) = \frac{P_{ini}}{E_b\pi r_b^2\delta}\left(\frac{e^{\delta x} + e^{-\delta x}}{e^{\delta L} - e^{-\delta L}}\right) \tag{71}$$

Then the nut is tightened and the pretensioning load results in the deflection of the plate. The redistribution of the stress occurs along the bolt, therefore the boundary conditions is changed to:

$$\begin{cases} S_S(L) = S_{ini}(L) - w \\ S'(0) = \varepsilon(0) = 0 \end{cases} \tag{72}$$

Solving Eq. (68) with the new boundary conditions gives:

$$S_S(x) = \left[\frac{P_{ini}}{E_b\pi r_b^2\delta}\left(\frac{e^{\delta L} + e^{-\delta L}}{e^{\delta L} - e^{-\delta L}}\right) - w\right]\left(\frac{e^{\delta x} + e^{-\delta x}}{e^{\delta L} + e^{-\delta L}}\right) \tag{73}$$

The distribution of shear stress and axial load along the bolt can be calculated as:

$$\tau(x) = k_{ini}\cdot S_S(x) = k_{ini}\left[\frac{P_{ini}}{E_b\pi r_b^2\delta}\left(\frac{e^{\delta L} + e^{-\delta L}}{e^{\delta L} - e^{-\delta L}}\right) - w\right]\left(\frac{e^{\delta x} + e^{-\delta x}}{e^{\delta L} + e^{-\delta L}}\right) \tag{74}$$

$$P_{rein}(x) = E_b\pi r_b^2\frac{dS_S(x)}{dx} = \left[P_{ini}\left(\frac{e^{\delta L} + e^{-\delta L}}{e^{\delta L} - e^{-\delta L}}\right) - E_b\pi r_b^2\delta w\right]\left(\frac{e^{\delta x} - e^{-\delta x}}{e^{\delta L} + e^{-\delta L}}\right) \tag{75}$$

Eq. (74) and Eq. (75) describe the stress distribution caused by the initial pretension. However, the continuing excavation process will result in the further radial deformation of the rock mass, and it will generate additional stress along the bolt. The stress distribution in this case can be calculated by the model proposed in Section 3.1.2.

An in situ test of a 25 mm diameter grouted bolt was carried out by Freeman[9] in the Kielder experimental tunnel. The tested sections of the tunnel was excavated by drill and blast. The bolt was pretensioned at 0.7 days and the advancement of the tunnel face was restarted at 0.9 days, which means the bolt at the installation moment can be only influenced by the initial pretension. So this case is suitable for the validation of the proposed model in this section. The parameters needed for the proposed model are adopted as follows. Length of the bolt, $L = 1.7$ m; Radius of the bolt, $r_b = 12.5$ mm; Young's modulus of the bolt, $E_b = 210$ GPa; Area of the face plate, $l_b \times l_b = 12\text{cm} \times 12\text{cm}$; Thickness of the face plate, $h = 2$ cm; Poisson's ratio of the face plate, $v = 0.25$; Equivalent stiffness of the face plate, $K_d = 320$ MN/m; and Initial shear stiffness at the bolt-grout interface, $K_{ini} = 2$ MPa/mm. It should be mentioned that $h$ and $K_{ini}$ are assumed typical value, and the equivalent stiffness of the plate $K_d$ was firstly calculated in 31. If the face plate is flat, $K_d$ can also be transformed to equivalent reaction coefficient $q = 28$ MN/mm by Eq. (56). The calculation results are shown in Fig. 15(a) and (b). The measured data are in accordance with the curves when the initial pretensioned load is set to 15 kN. From Eq. (54), the coefficient $\varphi$ under different $P_{ini}$ is calculated to be 0.63, which indicates that the loss of the prestress is proportional to the initial pretension load. This result coincides with the basic assumption in 31. Moreover, in some cases, the value of the plate deflection $w$ may exceed the value of $S_{ini}$ in Eq. (55). It will result in negative displacement and shear stress on the segment of the bolt near the excavation boundary, which is similar to the stress distribution pattern discussed in Section 3.1.2.

## 5. Limitations

Although analytical models of fully grouted bolts in pull-out tests and in situ rock masses are presented, there are still some limitations.

The fully grouted bolts discussed in this research mainly corresponds to cement grouted bolts. However, other grouting techniques are not covered in this research, such as resin grouting. Especially when the two-speed (fast and low) resin cartridges[30] are used, the load builds

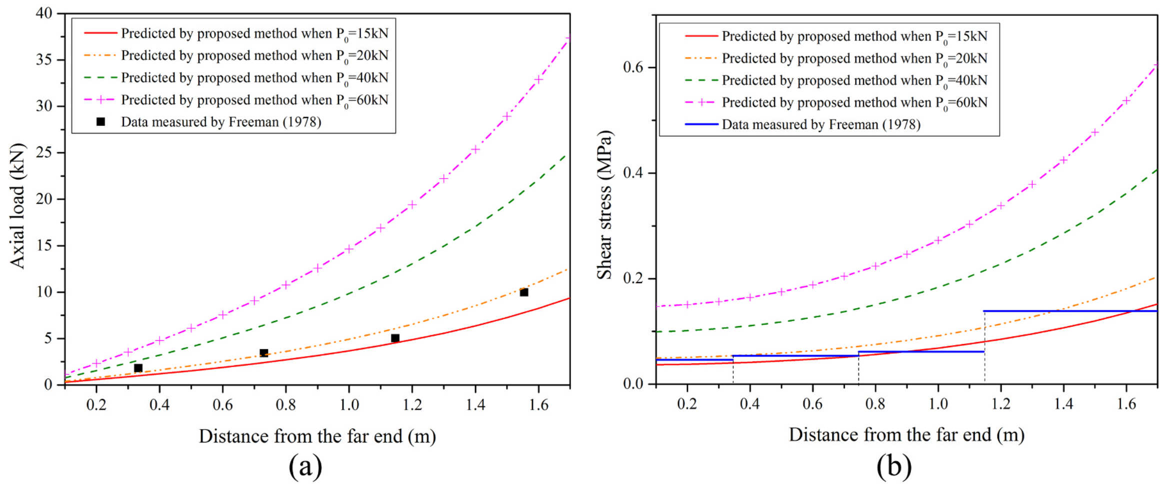


**Fig. 15.** Calculation results of in situ rock bolts (measured by 9). (a) Distribution of the axial load. (a) Distribution of the shear stress.

up along the bolt during the installation procedure. In this case, the bolt-grout (or grout-rock) bond properties are different from those discussed in this study.

For in-situ rock bolts, factors such as joints of the rock mass and installation of the adjacent bolts may also impact the reinforcement effect of bolt. In such cases, the proposed model needs further modifications.

## 6. Conclusions

This study presents new analytical models for fully grouted bolts in pull-out tests and in situ conditions. The stress distribution and load-displacement relationship are simulated by these models, and the predictions are demonstrated to be mathematically reasonable and show good agreement with the results of laboratory tests and field monitoring. Compared with the previous solutions, the following improvements have been achieved.

A new dynamic bond-slip model for fully grouted bolts under axial tension has been developed and can reflect the characteristic that the bond-slip relationship varies along the whole bond length of the bolt. In previous studies, the bond-slip relationship for fully grouted bolts was assumed to be constant.

The stress distribution, strain distribution and load-slip relationship of the bolts in pull-out tests are captured based on the dynamic bond-slip model. In addition, the displacement of the free end of the bolt is taken in account in the corresponding analytical solutions; previously, this displacement was often assumed to be zero.

A new analytical model is proposed for in situ rock bolts considering the non-homogeneous deformation of the rock and the interaction between the bolt and surrounding rock mass. This approach is applied to two bolt end scenarios for in situ bolts installed in rock by changing the boundary conditions.

A new analytical approach is presented to analyze the effect and loss of pretensioning force for the in situ pretensioned bolt with consideration of face plate deformation.

The new analytical model could not only provide a simpler and more explicit approach to understanding the mechanical behavior of the fully grouted bolts in pull-out tests and in situ but also offer an approach for the quantitative analysis of the reinforcement effect of the bolting system. Moreover, the proposed approach can also be embedded in numerical simulation software or other software packages to achieve deeper insights and study a broader range of applications.

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## References

1. Windsor CR. Rock reinforcement systems. *Int J Rock Mech Min Sci.* 1997;34:919–951.
2. Farmer IW. Stress distribution along a resin grouted rock anchor. *Int J Rock Mech Min Sci Geomech Abstr.* 1975;12:347–351.
3. Li C, Stillborg B. Analytical models for rock bolts. *Int J Rock Mech Min Sci.* 1999;36:1013–1029.
4. Ren FF, Yang ZJ, Chen JF, Chen WW. An analytical analysis of the full-range behaviour of grouted rockbolts based on a tri-linear bond-slip model. *Constr Build Mater.* 2010;24(3):361–370.
5. Martín LB, Tijani M, Hadj-Hassen F. A new analytical solution to the mechanical behaviour of fully grouted rockbolts subjected to pull-out tests. *Constr Build Mater.* 2011;25(2):749–755.
6. Ma S, Nemcik J, Aziz N. An analytical model of fully grouted rock bolts subjected to tensile load. *Constr Build Mater.* 2013;49(1):519–526.
7. Liu SS, Zhao MH, Li L. Analysis of dynamic broken line model based on experimental data of anchor anti-pullout. *Rock Soil Mech.* 2014;35(12):3389–3395.
8. Zou JF, Qian ZH. Face stability analysis of tunnel excavated below groundwater considering coupled flow-deformation. *Int J Geomech.* 2018;18(8):04018089.
9. Freeman TJ. The behavior of fully bonded rock bolts in the Kielder Experimental tunnel. *Tunn Tunn Int.* 1978;10(5):37–40.
10. Björnfot F, Stephansson O. Interaction of grouted rock bolts and hard rock masses at variable loading in a test drift of the Kiirunavaara Mine, Sweden. In: Proceedings of the International Symposium on Rock Bolting. Rotterdam: Balkema; 1984:337–395.
11. Cai Y, Esaki T, Jiang Y. An analytical model to predict axial load in grouted rock bolt for soft rock tunnelling. *Tunn Undergr Space Technol.* 2004;19(6):607–618.
12. Cai Y, Esaki T, Jiang Y. A rock bolt and rock mass interaction model. *Int J Rock Mech Min Sci.* 2004;41(7):1055–1067.
13. Cai Y, Jiang Y, Djamaluddin I, Iura T, Esaki T. An analytical model considering interaction behavior of grouted rock bolts for convergence–confinement method in tunneling design. *Int J Rock Mech Min Sci.* 2015;76:112–126.
14. Hyett AJ, Moosavi M, Bawden WF. Load distribution along fully grouted bolts, with emphasis on cable bolt reinforcement. *Int J Numer Anal Methods Geomech.* 1996;20(7):517–544.
15. Moosavi M, Grayeli R. A model for cable bolt-rock mass interaction: integration with discontinuous deformation analysis (DDA) algorithm. *Int J Numer Anal Methods Geomech.* 2006;43(4):661–670.
16. Nie W, Zhao ZY, Ning YJ, Sun JP. Development of rock bolt elements in two-dimensional discontinuous deformation analysis. *Rock Mech Rock Eng.* 2014;7(6):2157–2170.
17. Ma S, Zhao Z, Nie W, Gui Y. A numerical model of fully grouted bolts considering the tri-linear shear bond–slip model. *Tunn Undergr Space Technol.* 2016;54(4):73–80.
18. Li CC. Performance of D-bolts under static loading. *Rock Mech Rock Eng.* 2012;45(2):183–192.
19. Yazici S, Kaiser PK. Bond strength of grouted cable bolts. *Int J Rock Mech Min Sci Geomech Abstr.* 1992;29(3):279–292.
20. Kilic A, Yasar E, Celik AG. Effect of grout properties on the pull-out load capacity of

fully grouted rock bolt. *Tunn Undergr Space Technol.* 2002;17(4):355–362.
21. Benmokrane B, Chennouf A, Mitri HS. Laboratory evaluation of cement-based grouts and grouted rock anchors. *Int J Rock Mech Min Sci Geomech Abstr.* 1995;32(7):633–642.
22. Dai J, Ueda T, Sato Y. Unified analytical approaches for determining shear bond characteristics of FRP-concrete interfaces through pullout tests. *J Adv Conc Technol.* 2006;4(1):133–145.
23. Zhou YW, Wu YF, Yun Y. Analytical modeling of the bond–slip relationship at FRP-concrete interfaces for adhesively-bonded joints. *Compos Pt B-Eng.* 2010;41(6):423–433.
24. Liu SS, Zhao MH, Li L. Analysis of dynamic broken line model based on experimental data of anchor anti-pullout. *Rock Soil Mech.* 2014;35(12):3389–3395.
25. Rong G, Zhu HC, Zhou C. Testing study on working mechanism of fully grouted bolts of thread steel and smooth steel. *Chin J Rock Mech Eng.* 2004;23(3):469–475.
26. Carranza-Torres C, Fairhurst C. Application of the convergence-confinement method of tunnel design to rock masses that satisfy the Hoek-Brown failure criterion. *Tunn Undergr Space Technol.* 2000;15(2):187–213.
27. Zou JF, Chen KF, Pan QJ. An improved numerical approach in surrounding rock incorporating rockbolt effectiveness and seepage force. *Acta Geotech.* 2018;13(3):707–727.
28. Zou JF, Liu L, Xia MY A new simple approach for the quasi-plane strain problem of circular tunnel in strain-softening rock mass incorporating the effect of out-of-plane stress. Acta Geotech. DOI: 10.1007/s11440-018-0724-8.
29. Tao Z, Chen JX. Behaviour of rock bolting as tunnelling support. In: Stephansson O, editor. Proceedings of the International Symposium on Rock Bolting. Rotterdam: Balkema; 1984. p. 87-92.
30. Li CC. *Rockbolting: Principles and Applications.* Butterworth-Heinemann; 2017.
31. Ranjbarnia M, Fahimifar A, Oreste P. Practical method for the design of pretensioned fully grouted rockbolts in tunnels. *Int J Geomech.* 2016;16(1):04015012.
32. Ranjbarnia M, Fahimifar A, Oreste P. A simplified model to study the behavior of pre-tensioned fully grouted bolts around tunnels and to analyze the more important influencing parameters. *J Min Sci.* 2014;50(3):533–548.
33. Timoshenko S, Woinowsky S. *Theory of Plates and Shells.* McGraw-Hill; 1959:342–346.
34. Dym CL, Shames IH. *Solid Mechanics: A Variational Approach*. Augmented ed. New York: Springer; 2013:346–351.
35. Tan JY. Remarks of some problems for rectangular thin plates with free edges on elastic foundations. *Appl Math Mech.* 1991;12(4):415–420.